Kravchuk Encyclopedia
A Working Book Skeleton



Mikhail Kravchuk (1892–1942)

Portrait source: MacTutor History of Mathematics Archive,
University of St Andrews.

Editorial Preface

This editable skeleton follows the eighteen main topics and appendix program in Section 6.1 of the supplied article source, `KravFeinsKocik.tex`. Additional headings from the supplied archived website, `Kravchuk.zip`, have been placed within the closest corresponding chapters. The archived introduction now appears as the first numbered chapter; the eighteen topical chapters follow it. The contents preserve the original program rather than impose a new textbook sequence.

This remains a working encyclopedia, not a completed literature review. Developed sections now include finite matrix identities, source-based representation material, martingales, orthogonality, and Gaussian–Krawtchouk summation. Remaining gaps carry visible uppercase NEEDS TO BE DONE notices, including narrower unfinished tasks within otherwise developed sections. Spectral-asymptotic working material remains explicitly distinguished from proved identities. The source register below separates supplied materials, editorial derivations, and bibliography records awaiting independent verification.

The title retains "Kravchuk" while the polynomial notation uses the conventional spelling "Krawtchouk." Historical names and original terminology should be checked as chapters are developed. In particular, the operator symbols A, K, Λ, and T require definitions from their original source contexts before they are used interchangeably.

Notation and Conventions

Throughout the starter material, $K_n(j;N)$ is defined by the ordinary generating function, $k_n(N-2j,N) = n!K_n(j;N)$, $\Phi_N = (K_n(j;N))$, $B_N = \mathrm{diag}(\binom{N}{j})$, and $S_N = \Phi_N B_N$. All matrix indices begin at zero. In the standard column-vector basis, $R_N e_j = (N-j)e_{j+1}$, $L_N e_j = j e_{j-1}$, and $A_N = R_N + L_N$. The diagonal matrix Λ_N has entries $N-2j$. These conventions, including the distinction between Φ_N and S_N, are developed in Sections 1, 3.1, and 1. Appendix A contains verified examples for $N = 2,3,4,5,7,8$. Physicists' and probabilists' Hermite normalizations must be explicitly distinguished. Binomial coefficients outside their usual integer range vanish.

Dictionary for the integrated Kocik article

The notation of [51] is translated rather than adopted. Its total order n, row p and column q become our N, n, j; its Krawtchouk matrix K becomes Φ_N, its Kac matrix M becomes A_N, and its eigenvalue matrix becomes Λ_N. The source's diagonal binomial metric Γ becomes B_N; its Pascal matrix B becomes the already established P_N. The source's operator $N = [L, R]$ becomes Λ_N on the fixed-order space, not the scalar order N. We use the ordinary commutator $[X, Y] = XY - YX$, distinguishing it from the source's half-commutator. The symbols Φ_N, $S_N = \Phi_N B_N$, and U_N retain their distinct meanings throughout. The parity matrix is D_N; powers-of-two weights in Pascal coordinates are W_N.

Preferred conventions for the newly supplied summary papers

For general-parameter transform material we retain the supplied `KravPQ.tex` notation $K_n(x, N)$ with exponential coefficients. For the sums material we retain ϕ_{nj}^N and its symmetric specialization Φ_{nj}^N from `KravSums.tex`. Chapter 20 supplies the parameter interchange, factorial scaling, and dictionary to earlier chapters. Previously established symmetric notation is retained there to avoid silently altering formulas.

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CHAPTER 1

Introduction

`Omnibus ex nihilo ducendis sufficit unum.`
"To make all things from nothing, unity suffices."
Leibniz

The Kravchuk Encyclopedia
Ph. Feinsilver and J. Kocik

So it is written on a medal entitled *Imago Creationis* and designed by Leibniz to *"exhibit to posterity in silver"* his discovery of the binary system.

Note on spellings. Since the original articles appeared in the French Academy journal *C.R.A.S.*, the French spelling Krawtchouk was used and has been generally adopted. We will use with equal propriety the English spelling Kravchuk, which, in fact, we encourage in the current context.

This is the beginnings of a 'Kravchuk Encyclopedia'. From the outline presented here, as the project develops, the gaps will gradually be filled in. We hope that you will find lots of interesting material in this work.

1. Mikhail Kravchuk (1892–1942): life and mathematical legacy

Mikhail Kravchuk, also known as Mykhailo Pylypovych Kravchuk, was a Ukrainian mathematician whose work ranged across algebra, differential and integral equations, probability, and mathematical statistics. He was born on 27 September 1892 in Chovnitsy, in Volyn, in what is now Ukraine, and died on 9 March 1942 in a Soviet forced-labour camp in Kolyma. The biographical outline here draws on the MacTutor History of Mathematics Archive and the institutional biography maintained by the Institute of Mathematics of the National Academy of Sciences of Ukraine.[1]

Education and scientific career. Kravchuk's family moved to Lutsk in 1901. He completed his schooling there with a gold medal in 1910 and entered the University of Kyiv, then St Vladimir University, graduating in 1914. His early research included commuting matrices. After the upheavals of the First World War and the subsequent revolutionary and civil-war years,

[1]MacTutor History of Mathematics Archive, University of St Andrews, "Mykhailo Pilipovich Krawtchouk," https://mathshistory.st-andrews.ac.uk/Biographies/Krawtchouk/; Institute of Mathematics, National Academy of Sciences of Ukraine, "Kravchuk Mikhail Pilipovich (1892–1942)," https://www.imath.kiev.ua/famous?n=kravchuk&lang=en. These are secondary biographical accounts, not archival documents.

he combined university teaching with work in schools. He defended his doctoral dissertation in 1924 and received the title of professor in 1925. His work on matrices, bilinear forms, and linear transformations was an important part of this early development.

In 1929 he was elected a full member of the Ukrainian Academy of Sciences. He taught at the Kyiv Polytechnic Institute and served as head of its mathematics chair. The Institute of Mathematics identifies him as one of its founders. His activity was not confined to research: he helped develop Ukrainian-language mathematical education and terminology, wrote teaching materials, and supported the training of younger mathematicians.

The polynomials and the scope of his work. The orthogonal polynomials bearing his name were introduced in 1929. They are associated with the binomial distribution, providing the historical starting point for the finite orthogonality and transform identities developed in this Encyclopedia. Their subsequent uses in coding theory, finite transforms, and algebraic constructions belong to the later development of the subject; they should not all be attributed to Kravchuk himself.

Although these polynomials are now his most familiar mathematical legacy, his research was considerably broader. It included matrix theory, the method of moments, differential and integral equations, probability, statistics, and questions in mathematical education. This wider setting is particularly relevant here: the interplay between matrices, orthogonality, and operator methods is not merely an accidental grouping of topics under one name.

Persecution, death, and rehabilitation. Kravchuk was arrested in 1938 during Stalin's purges and sentenced on political charges to twenty years of imprisonment. He was sent to Kolyma, where he died in 1942, aged forty-nine. MacTutor describes coercion and threats against his family during the proceedings. He was posthumously rehabilitated in 1956. His membership in the Ukrainian Academy of Sciences was restored in 1992. These events are essential to an account of his career: its abrupt end was a consequence of political repression, not a natural conclusion to his scientific work.[2]

Names and conventions in this Encyclopedia. The spellings *Kravchuk*, *Krawtchouk*, and *Krawtschuk* reflect different transliterations. MacTutor notes that *Krawtchouk* was the spelling used in his French-language papers. We retain *Kravchuk* in the Encyclopedia's title and historical discussion, while respecting *Krawtchouk* in established mathematical terminology, source titles, and quotations. The first names *Mikhail* and *Mykhailo* likewise occur in the biographical literature. These variants refer to the same mathematician.

[2]For the account of his persecution, rehabilitation, and restoration of Academy membership, see the MacTutor biography cited above.

Editorial provenance. This sourced biographical section is a later addition to the Encyclopedia. It is not part of the archived 2001 introduction, whose wording and attribution are retained separately.

Starting points

The original introduction offered the following entry points, now represented within this book:

- *First View*: see Section 5.
- *Krawtchouk Matrices*: see Section 4.
- *Kravchuk Polynomials*: see Section 2.
- *Twistons*: see Section 6.
- *Other Examples*: reserved for future additions to the introductory material.

For transforms, see Chapter 15. The complete table of contents replaces the original website's *To Contents* link.

Archival note

This chapter transcribes the substantive text of `Kravchuk/Introduction.html` in the supplied website archive. The displayed author attribution and the historical wording above are retained from that source; they do not assign authorship of later editorial additions. Its modification footer records July 28, 2001. The historical claims and quotation are reproduced as source text and have not been independently verified for this skeleton.

The original page also linked to an interactive K-transform form and a bibliographic search database. Those were website services rather than book text; they are recorded here as archival resources, not represented as functioning services. Navigation has been adapted to book cross-references.

CHAPTER 2

Pascal's Triangle

1. Pascal's triangle and binomial identities

Choosing an n-element subset of $\{1,\dots,N\}$ according to whether it contains the last element proves

$$\binom{N}{n} = \binom{N-1}{n} + \binom{N-1}{n-1}. \tag{1}$$

We use $\binom{N}{n} = 0$ outside $0 \le n \le N$. Expanding a product of N factors gives

$$(1+v)^N = \sum_{n=0}^{N} \binom{N}{n} v^n.$$

Taking coefficients in $(1+v)^{a+b} = (1+v)^a(1+v)^b$ gives Vandermonde's identity

$$\binom{a+b}{n} = \sum_r \binom{a}{r}\binom{b}{n-r}.$$

Replacing the second factor by $(1-v)^b$ yields the signed convolution defining $K_n(b; a+b)$ in Section 2. Thus the first column of the Krawtchouk array is Pascal's row, and its other columns are signed subset counts, not probability distributions. For example, at $N = 3$, the columns for $j = 0$ and $j = 1$ are respectively $(1,3,3,1)^T$ and $(1,1,-1,-1)^T$.

2. Krawtchouk polynomials

For $0 \le n, j \le N$, define the symmetric Krawtchouk polynomial values by

$$(1+v)^{N-j}(1-v)^j = \sum_{n=0}^{N} K_n(j;N) v^n. \tag{2}$$

Hence

$$K_n(j;N) = \sum_r (-1)^r \binom{j}{r}\binom{N-j}{n-r}.$$

Binomial coefficients outside their usual range are understood to vanish. In centered coordinates $x = N - 2j$, write

$$(1+v)^{(N+x)/2}(1-v)^{(N-x)/2} = \sum_{n=0}^{N} \frac{v^n}{n!} k_n(x,N).$$

Thus $k_n(N-2j,N) = n!K_n(j;N)$.

3. Generating functions

Write $x = N - 2j$ and use the generating function in (2). Logarithmic differentiation gives the polynomial identity

$$(1-v^2)\frac{\partial G_{N,j}}{\partial v} = (x - Nv)G_{N,j}.$$

Coefficient comparison, with $K_{-1} = 0$, yields

$$(n+1)K_{n+1}(j;N) = xK_n(j;N) - (N-n+1)K_{n-1}(j;N).$$

In particular,

$$K_0 = 1, \qquad K_1 = x, \qquad K_2 = \frac{x^2 - N}{2}, \qquad K_3 = \frac{x^3 - (3N-2)x}{6}.$$

The factorial-scaled polynomials are $k_n = n!K_n$, so their leading coefficient as polynomials in x is one. This is why factorials must be retained when passing between ordinary and exponential generating functions. At the endpoints,

$$K_n(0;N) = \binom{N}{n}, \qquad K_n(N;N) = (-1)^n\binom{N}{n}.$$

Moreover, $G_{N,N-j}(v) = G_{N,j}(-v)$, giving $K_n(N-j;N) = (-1)^n K_n(j;N)$. These are fixed-order identities, distinct from the cross-order construction of Section 5.

4. Krawtchouk matrices

Set $\Phi_N = (K_n(j;N))_{n,j=0}^N$ and $B_N = \operatorname{diag}(\binom{N}{0},\dots,\binom{N}{N})$. Binomial duality is

$$\binom{N}{j}K_n(j;N) = \binom{N}{n}K_j(n;N).$$

Consequently $S_N = \Phi_N B_N$ is symmetric. The symmetric transform satisfies

$$\Phi_N^2 = 2^N I.$$

For orientation,

$$\Phi_1 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad \Phi_2 = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}.$$

5. First view and cross properties

Write $G_{N,j}(v) = (1+v)^{N-j}(1-v)^j$, as in (2). Multiplying by $1+v$ or $1-v$ and comparing coefficients gives

$$K_n(j; N+1) = K_n(j; N) + K_{n-1}(j; N), \tag{3}$$

$$K_n(j+1; N+1) = K_n(j; N) - K_{n-1}(j; N). \tag{4}$$

Here $0 \le j \le N$, and coefficients outside $0 \le n \le N$ vanish at order N. Adding and subtracting yields the inverse cross rules

$$K_n(j; N+1) + K_n(j+1; N+1) = 2K_n(j; N),$$
$$K_n(j; N+1) - K_n(j+1; N+1) = 2K_{n-1}(j; N).$$

These are cross-order identities: the order parameter must not be suppressed when they are used to construct successive tables. For example, $K_2(1;3) = -1$ and $K_1(1;3) = 1$ give $K_2(1;4) = 0$ and $K_2(2;4) = -2$.

There is also a fixed-order local relation. If

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} K_n(j;N) & K_n(j+1;N) \\ K_{n+1}(j;N) & K_{n+1}(j+1;N) \end{pmatrix},$$

then $-a+c = b+d$. Indeed, $(1-v)G_{N,j} = (1+v)G_{N,j+1}$. Column reversal and row reversal are instead the fixed-order symmetries proved in Section 1; these should be distinguished from the cross-order rules above. Compare [**51**, Appendix C].

CHAPTER 3

Random Walks

1. Paths and path integrals

Consider length-N binary paths $w = (w_1, \dots, w_N)$, where $w_a \in \{0, 1\}$ and $|w| = \sum_a w_a$ is the endpoint coordinate. Fix j distinguished steps and assign the signed weight

$$\omega_j(w) = (-1)^{w_1 + \cdots + w_j}.$$

The particular first-j choice is inessential; any fixed subset of j steps gives the same endpoint sums. A path with endpoint n and r right steps among the distinguished steps has weight $(-1)^r$. There are $\binom{j}{r}\binom{N-j}{n-r}$ such paths, so

$$\sum_{|w|=n} \omega_j(w) = \sum_r (-1)^r \binom{j}{r}\binom{N-j}{n-r} = K_n(j; N). \tag{5}$$

This is a path interpretation of the coefficient formula in Section 2, not a new definition. For $j = 0$ it counts ordinary paths and gives $\binom{N}{n}$; for $j > 0$, cancellation produces signed entries of column j of Φ_N. For instance, $N = 4$, $j = 2$ gives endpoint sums $(1, 0, -2, 0, 1)$.

The signed Galton-board construction of [**51**, Section 6] is an elementary analogue of a sum over paths. Its weights are not probabilities. In quantum mechanics one sums amplitudes before taking an appropriately normalized absolute square; a physical unitary model requires more structure than this combinatorial identity. The unsigned walk and the stochastic Ehrenfest transition operator should likewise not be identified with a unitary evolution.

1.1. Weighted paths and the two-parameter deformation.

Over a commutative ring, define a separate coefficient array

$$\mathcal{K}_n(j; N; \alpha, \beta) = [v^n](1 + \alpha v)^{N-j}(1 + \beta v)^j.$$

A consistent path rule gives every left step weight 1, and a right step weight β at each of j distinguished times and α at all other times. Thus

$$\mathcal{K}_n(j; N; \alpha, \beta) = \sum_r \binom{j}{r}\binom{N-j}{n-r}\beta^r \alpha^{n-r}.$$

This recovers $K_n(j;N)$ at $(\alpha,\beta)=(1,-1)$ without changing its normalization. Coefficient comparison gives

$$\mathcal{K}_n(j;N+1)=\mathcal{K}_n(j;N)+\alpha\mathcal{K}_{n-1}(j;N),$$
$$\mathcal{K}_n(j+1;N+1)=\mathcal{K}_n(j;N)+\beta\mathcal{K}_{n-1}(j;N).$$

For adjacent entries $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ in this array, $\beta a+c=\alpha b+d$. These identities require no division by $\alpha-\beta$ and remain valid when parameters coincide or are not invertible.

REMARK 3.1 (A discrepancy in the source's path rule). The displayed generating function of [51, Section 6.2] is the one used here. However, the preceding prose assigns left/right weights α,β only at the j distinguished times, and weights $1,1$ elsewhere. That different rule produces $(1+v)^{N-j}(\alpha+\beta v)^j$, not the displayed generating function. We retain the displayed coefficient array and specify the compatible rule explicitly. No stochastic normalization, orthogonality, or involution property is asserted for arbitrary parameters.

2. A, K, and Λ

For the symmetric walk on $\{0,\dots,N\}$, use column vectors for state distributions and the zero-based standard basis e_j. Put

$$A_Ne_j=(N-j)e_{j+1}+je_{j-1},\qquad \Lambda_N=\operatorname{diag}(N,N-2,\dots,-N),$$

with endpoint terms omitted. In this section the matrix symbol K is explicitly defined to mean $\Phi_N=(K_n(j;N))_{n,j=0}^N$. It is not a single polynomial K_n or the symmetric matrix $S_N=\Phi_NB_N$.

2.1. Intertwining and diagonalization. Writing $G_j(v)=(1+v)^{N-j}(1-v)^j$, differentiation gives

$$(1-v^2)G_j'(v)=(N-2j-Nv)G_j(v).$$

Equating coefficients yields

$$(N-n+1)K_{n-1}(j;N)+(n+1)K_{n+1}(j;N)=(N-2j)K_n(j;N).$$

The spatial recurrence is

$$jK_n(j-1;N)+(N-j)K_n(j+1;N)=(N-2n)K_n(j;N).$$

Absent endpoint terms are zero. In matrix form these identities read

$$A_N\Phi_N=\Phi_N\Lambda_N,\qquad \Phi_NA_N=\Lambda_N\Phi_N.$$

This is the coefficient proof of the master equation; its symmetric-power interpretation is given in Section 1. Since $\Phi_N^2=2^NI$, we obtain

$$\Phi_N^{-1}=2^{-N}\Phi_N,\qquad A_N=2^{-N}\Phi_N\Lambda_N\Phi_N.$$

Column j of Φ_N is a right eigenvector with eigenvalue $N-2j$; row n is a left eigenvector with eigenvalue $N-2n$.

2.2. Transition probabilities. For $N \geq 1$, $T_N = A_N/N$ is column-stochastic. Hence $p_{t+1} = T_N p_t$ and, for integers $t \geq 0$,

$$T_N^t = 2^{-N}\Phi_N(\Lambda_N/N)^t\Phi_N.$$

The backward operator on column vectors of function values is T_N^T, not T_N; its eigenfunctions are $j \mapsto K_n(j;N)$. The stationary column distribution is $2^{-N}(\binom{N}{0},\ldots,\binom{N}{N})^T$, the normalized first column of Φ_N. The eigenvalue -1 of T_N expresses period two; stationarity does not imply convergence from arbitrary initial data.

2.3. The cases $N = 3$ and $N = 5$. For $N = 3$,

$$A_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 3 & 0 & 2 & 0 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$\Lambda_3 = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}.$$

The transition eigenvalues are $1, \frac{1}{3}, -\frac{1}{3}, -1$. For $N = 5$,

$$A_5 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 5 & 0 & 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 2 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$\Lambda_5 = \begin{pmatrix} 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -5 \end{pmatrix}.$$

The transition eigenvalues are $1, \frac{3}{5}, \frac{1}{5}, -\frac{1}{5}, -\frac{3}{5}, -1$. The corresponding R_N, L_N, Φ_N and S_N tables are in Appendix A. Centered and lattice polynomial tables are in Appendix B.

3. Nonsymmetric walks

NONSYMMETRIC WALKS NEEDS TO BE DONE.

4. Symmetric Krawtchouk matrices and binomial expectations

SYMMETRIC KRAWTCHOUK MATRICES AND BINOMIAL EXPECTATIONS NEEDS TO BE DONE.

Connection to the supplied article sources

The supplied source `KravFeinsKocik.tex`, Section 5, develops the classical Bernoulli random walk and a multivariate extension. The supplied `jkhadamard_Oct_08_00.tex`, Section 5, connects classical random walks with Krawtchouk matrices. Read these alongside this chapter using the book's ordinary generating-function normalization; source conventions should not be silently substituted for it.

CHAPTER 4

Urn Model

1. Markov chains

In the discrete-time Ehrenfest chain, the state j is the number of balls in one of two urns. Moving one uniformly chosen ball gives

$$(Pf)(j) = \frac{N-j}{N} f(j+1) + \frac{j}{N} f(j-1).$$

The Krawtchouk difference equation gives

$$PK_n(\cdot; N) = \left(1 - \frac{2n}{N}\right) K_n(\cdot; N).$$

Here $N \geq 1$, and terms outside the state space have zero transition coefficient.

2. Initial and invariant distributions

The invariant distribution is $w_N(j) = 2^{-N}\binom{N}{j}$. Detailed balance follows from

$$w_N(j)\frac{N-j}{N} = w_N(j+1)\frac{j+1}{N}.$$

The non-lazy discrete-time chain has period two; invariance should not be confused with convergence to equilibrium from every initial distribution.

Connection to the supplied article sources

Section 4 ("Ehrenfest urn model") of the supplied `KravFeinsKocik.tex` belongs to this chapter. The urn state is the Hamming weight of a binary configuration: flipping a uniformly selected coordinate moves from j to $j+1$ with probability $(N-j)/N$ and to $j-1$ with probability j/N, for $N \geq 1$. In the book's column-vector convention the transition matrix is A_N/N. The tensor-space realization is discussed in Chapter 10.

CHAPTER 5

Symmetric Functions and Energy

1. Elementary symmetric functions and determinants

For scalars $z_1, \dots, z_N$, the elementary symmetric functions are

$$e_n(z_1, \dots, z_N) = \sum_{1 \le a_1 < \cdots < a_n \le N} z_{a_1} \cdots z_{a_n}, \qquad e_0 = 1.$$

Each subset occurs once: repeated indices and ordered tuples are excluded. For $D = \operatorname{diag}(z_1, \dots, z_N)$,

$$\det(I + vD) = \prod_{a=1}^{N} (1 + z_a v) = \sum_{n=0}^{N} e_n(z_1, \dots, z_N) v^n. \tag{6}$$

If $z_a = \epsilon_a \in \{+1, -1\}$ and precisely j signs are negative, comparison with the defining generating function (2) gives

$$e_n(\epsilon_1, \dots, \epsilon_N) = K_n(j; N). \tag{7}$$

Thus the order-n interaction energy depends only on the number of negative signs, not their positions. The first two nonconstant energies are

$$e_1 = N - 2j, \qquad e_2 = \frac{(N-2j)^2 - N}{2}.$$

These are ordinary coefficient values, not the factorial-scaled centered polynomials: $k_n(N - 2j, N) = n! e_n$. The twiston interpretation is given in Section 6; the signed-path interpretation in Section 1 gives the same coefficients by a different combinatorial construction. See also [**51**, Sections 1 and 5].

2. Traces on Grassmann algebras

Let V have dimension N and let $T : V \to V$ be linear. On the nth exterior power define

$$(\bigwedge^{n} T)(v_1 \wedge \cdots \wedge v_n) = Tv_1 \wedge \cdots \wedge Tv_n.$$

The trace of this induced map is the sum of the principal $n \times n$ minors of T. Consequently,

$$\det(I + vT) = \sum_{n=0}^{N} v^n \operatorname{tr}(\bigwedge^{n} T). \tag{8}$$

Indeed, in an exterior basis indexed by subsets, the diagonal entry indexed by a subset is precisely its principal minor; the determinant expansion gives the same coefficients. This argument does not require diagonalizability. For a sign operator D with $N-j$ eigenvalues $+1$ and j eigenvalues -1, the degree-n trace is therefore

$$\operatorname{tr}(\bigwedge^{n} D) = K_n(j;N).$$

For the full Grassmann algebra $\bigwedge V = \bigoplus_n \bigwedge^n V$, ordinary trace and supertrace are respectively

$$\operatorname{tr}(\bigwedge T) = \det(I+T), \qquad \operatorname{str}(\bigwedge T) = \sum_n (-1)^n \operatorname{tr}(\bigwedge^{n} T) = \det(I-T).$$

Here $\bigwedge^0 T$ is the identity on the scalar space. These formulas concern multiplicative exterior powers, not the additive Lie-algebra action obtained by applying a generator to one wedge factor at a time. They develop the Grassmann-algebra heading of the supplied source outline from the determinant identity in Section 1.

CHAPTER 6

Martingales

1. Iterated integrals

Let $\epsilon_1, \epsilon_2, \dots$ be independent symmetric signs, $X_m = \sum_{a=1}^{m} \epsilon_a$, and $\mathcal{F}_m = \sigma(\epsilon_1, \dots, \epsilon_m)$. Define

$$M_m(v) = \prod_{a=1}^{m} (1 + v\epsilon_a) = \sum_{n=0}^{m} E_n(m) v^n.$$

Independence and zero mean imply $\mathbb{E}[M_{m+1}(v) \mid \mathcal{F}_m] = M_m(v)$. Comparing coefficients gives a martingale in m for every fixed n, taking $E_n(m) = 0$ for $n > m$. Direct expansion gives

$$E_n(m) = \sum_{1 \le a_1 < \cdots < a_n \le m} \epsilon_{a_1} \cdots \epsilon_{a_n} = \frac{k_n(X_m, m)}{n!}. \tag{9}$$

With $E_0(m) = 1$, the discrete stochastic integral recursion is

$$E_n(m+1) - E_n(m) = \epsilon_{m+1} E_{n-1}(m), \qquad E_n(m) = \sum_{a=1}^{m} E_{n-1}(a-1)\epsilon_a. \tag{10}$$

The integrand is evaluated before the new sign is observed. Thus these are iterated sums over strictly increasing times, with no diagonal terms. For example, $E_1(m) = X_m$ and $E_2(m) = (X_m^2 - m)/2$. This develops the iterated-integral interpretation in the probabilistic part of the supplied `KravFeinsKocik.tex`; all normalizations here follow (2).

2. Orthogonal functionals

For fixed N, independence gives the two-variable generating identity

$$\mathbb{E}[M_N(v) M_N(w)] = \prod_{a=1}^{N} \mathbb{E}[(1 + v\epsilon_a)(1 + w\epsilon_a)] = (1 + vw)^N.$$

Consequently,

$$\mathbb{E}[E_n(N) E_m(N)] = \delta_{nm} \binom{N}{n}. \tag{11}$$

Equivalently, if $J = (N - X_N)/2$ has mass $\pi_N(j) = 2^{-N}\binom{N}{j}$, then

$$\sum_{j=0}^{N} \pi_N(j) K_n(j;N) K_m(j;N) = \delta_{nm} \binom{N}{n}.$$

This also follows by expanding into subsets: two distinct subsets leave at least one independent sign to an odd power, so their product has expectation zero. Identical subsets contribute one. There are $N+1$ nonzero orthogonal functions on the $N+1$-point lattice, so they form a basis of all functions of J. Explicitly,

$$f(j) = \sum_{n=0}^{N} \widehat{f}_n K_n(j;N), \qquad \widehat{f}_n = \frac{\mathbb{E}[f(J)K_n(J;N)]}{\binom{N}{n}}, \qquad \mathbb{E}[f(J)^2] = \sum_{n=0}^{N} \binom{N}{n} \widehat{f}_n^2$$

for real f. In factorial normalization the squared norm is $(n!)^2\binom{N}{n}$, not merely $\binom{N}{n}$. The martingale time parameter N should be distinguished from the fixed-order lattice argument j.

3. Krawtchouk polynomials and multinomial distribution

The supplied source extends the product martingale to independent identically distributed finite-valued variables. Let $\mathbb{P}(Y_a = \xi_r) = p_r$ for $0 \le r \le d-1$, with $p_r > 0$, $\sum_r p_r = 1$, mean μ, and variance $\sigma^2 > 0$. If ν_r counts occurrences of ξ_r among N observations, set

$$\mathcal{G}_{\boldsymbol{\nu}}(v) = \prod_{r=0}^{d-1} (1 + v(\xi_r - \mu))^{\nu_r} = \sum_{n=0}^{N} Q_n(\boldsymbol{\nu}) v^n.$$

Then

$$Q_n(\boldsymbol{\nu}) = \sum_{a_0+\cdots+a_{d-1}=n} \prod_{r=0}^{d-1} \binom{\nu_r}{a_r} (\xi_r - \mu)^{a_r}, \qquad Q_n(\boldsymbol{\nu}+\mathbf{e}_r) = Q_n(\boldsymbol{\nu}) + (\xi_r - \mu) Q_{n-1}(\boldsymbol{\nu}).$$

The count vector has multinomial mass $N! \prod_r p_r^{\nu_r}/\nu_r!$ on $\sum_r \nu_r = N$. Since a centered increment has mean zero and variance σ^2,

$$\mathbb{E}[\mathcal{G}_{\boldsymbol{\nu}}(v)\mathcal{G}_{\boldsymbol{\nu}}(w)] = (1 + \sigma^2 v w)^N, \qquad \mathbb{E}[Q_n Q_m] = \delta_{nm} \binom{N}{n} \sigma^{2n}.$$

The same conditional-expectation argument as in Section 1 proves the coefficient martingale property as observations are added.

For the two values $+1, -1$ with equal probabilities this reduces exactly to $Q_n = K_n(j;N)$. For $d > 2$, these $N+1$ orthogonal functions are generally not a complete basis on the $\binom{N+d-1}{d-1}$ count vectors. The source's scalar-parameter construction is therefore distinguished here from a full multi-index multivariate Krawtchouk system; the latter needs additional independent contrasts and is not asserted by this argument.

CHAPTER 7

Lie Algebras and Krawtchouk Polynomials

1. Finite raising and lowering matrices

Let $e_0, \dots, e_N$ be the standard column-vector basis of $\mathbb{R}^{N+1}$. Define

$$R_N e_j = (N-j)e_{j+1}, \qquad L_N e_j = j e_{j-1}, \qquad A_N = R_N + L_N, \tag{12}$$

with endpoint terms omitted, and put $\Lambda_N = \operatorname{diag}(N, N-2, \dots, -N)$. Equivalently, the only nonzero entries of R_N and L_N are

$$(R_N)_{j+1,j} = N - j, \qquad (L_N)_{j,j+1} = j + 1, \qquad 0 \le j < N.$$

Thus R_N raises the basis index, while L_N lowers it. Their sum is tridiagonal, but is not symmetric in this basis when $N > 1$. In general $R_N^T \neq L_N$; the normalized angular-momentum matrices use a different diagonal normalization.

Direct calculation gives

$$[L_N, R_N] = \Lambda_N, \qquad [\Lambda_N, R_N] = -2R_N, \qquad [\Lambda_N, L_N] = 2L_N. \tag{13}$$

These relations fix the sign convention for the finite $\mathfrak{sl}(2)$ representation. For example,

$$R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}, \quad L_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}, \quad A_2 = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}.$$

The full matrices for $N = 2, 3, 4, 5, 7, 8$ are in Appendix A. Their intertwining with the Krawtchouk matrix is explained in Section 1.1.

1.1. Differential-operator dictionary. Use independent auxiliary variables ξ, η for homogeneous polynomials, to avoid confusing them with the centered lattice variable $x = N - 2j$. On $e_j = \xi^{N-j}\eta^j$,

$$R = \eta\partial_\xi, \qquad L = \xi\partial_\eta, \qquad \mathcal{H} = \xi\partial_\xi - \eta\partial_\eta.$$

Their matrices are R_N, L_N, Λ_N, respectively. The Euler operator $\mathcal{E} = \xi\partial_\xi + \eta\partial_\eta$ acts as NI, and is not $\mathcal{H}$. For the group convention of Section 1, differentiation gives

$$d\rho_N \begin{pmatrix} a & b \\ c & d \end{pmatrix} = (a\xi + c\eta)\partial_\xi + (b\xi + d\eta)\partial_\eta.$$

In particular,

$$d\rho_N(F) = R + L, \qquad d\rho_N(G) = \mathcal{H}, \qquad d\rho_N(i) = L - R.$$

The last sign follows from the fixed matrix $i = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ and the transpose in the polynomial substitution. It is not $R - L$ in this convention. This dictionary reconciles the operator treatment in [51, Section 7.4] with the book's column-vector basis. The commutators are those of (13); no second normalization is needed.

2. $\mathfrak{so}(2,1)$ explained

The pure split quaternions form a real three-dimensional space

$$V = \{ti + xF + yG : t, x, y \in \mathbb{R}\}, \qquad Q(t,x,y) = t^2 - x^2 - y^2.$$

Here $i^2 = -1$, $F^2 = G^2 = 1$, $iF = G$, $FG = -i$, and $Gi = F$; distinct units anticommute. With the ordinary commutator $[X, Y] = XY - YX$,

$$[i, F] = 2G, \qquad [F, G] = -2i, \qquad [G, i] = 2F.$$

These are the structure constants of $\mathfrak{sl}(2,\mathbb{R})$ in a split-quaternion basis. The form Q is invariant under conjugation, so $X \mapsto \mathrm{ad}_X$, where $\mathrm{ad}_X(Y) = [X, Y]$, maps this algebra isomorphically onto $\mathfrak{so}(1,2)$. Changing the overall sign of the metric identifies the same algebra with the notation $\mathfrak{so}(2,1)$ used in this chapter.

For example, in the ordered basis (i, F, G) the infinitesimal actions are

$$\mathrm{ad}_i = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -2 \\ 0 & 2 & 0 \end{pmatrix}, \quad \mathrm{ad}_F = \begin{pmatrix} 0 & 0 & -2 \\ 0 & 0 & 0 \\ -2 & 0 & 0 \end{pmatrix}, \quad \mathrm{ad}_G = \begin{pmatrix} 0 & 2 & 0 \\ 2 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Each satisfies $M^T\eta + \eta M = 0$ for $\eta = \mathrm{diag}(1, -1, -1)$. The source uses $\frac{1}{2}(XY - YX)$ on pure quaternions; consequently its structure constants lack the factors of two displayed here. See [51, Section 7]; notation follows the book conventions.

3. $\mathfrak{so}(2,1)$ spinors

The real spinor space is $\mathbb{R}^2$. The split units act by

$$i = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad F = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad G = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.$$

The spinor action $u \mapsto gu$ of $g \in SL(2,\mathbb{R})$ is distinct from the vector action $X \mapsto gXg^{-1}$ on traceless matrices. The latter preserves $\det X$ and gives the double covering

$$SL(2,\mathbb{R}) \longrightarrow SO^+(1,2), \qquad \ker = \{I, -I\}.$$

The superscript $+$ specifies the identity component; this covering must not be asserted onto every component of $SO(1,2)$.

The compact generator gives $\exp(\theta i) = \cos\theta\, I + \sin\theta\, i$; the noncompact generator gives $\exp(tF) = \cosh t\, I + \sinh t\, F$. On the vector space, the adjoint action of $\exp(\theta i)$ rotates the (F, G) plane by 2θ, while $\exp(tF)$ produces a boost in the (i, G) plane with parameter $2t$ (with the signs fixed by the

commutators). This doubling explains why g and $-g$ have the same vector action. Higher symmetric powers of this two-dimensional spinor representation produce the finite Krawtchouk construction in Section 1. See [**51**, Section 7]; notation follows the book conventions.

4. Quaternions and Clifford algebras

The split quaternion algebra is

$$\mathbb{H}_s = \{a + bi + cF + dG : a, b, c, d \in \mathbb{R}\},$$

with the multiplication rules of Section 2. Define

$$\overline{q} = a - bi - cF - dG, \qquad \mathrm{N}(q) = q\overline{q} = a^2 + b^2 - c^2 - d^2.$$

Conjugation reverses products, and the norm is multiplicative. Thus $q^{-1} = \overline{q}/\mathrm{N}(q)$ precisely when $\mathrm{N}(q) \neq 0$. Unlike ordinary quaternions, this algebra contains nonzero zero divisors, for example $(1+F)(1-F) = 0$. The matrix realization is

$$q \longmapsto \begin{pmatrix} a+d & b+c \\ -b+c & a-d \end{pmatrix}, \qquad \det q = \mathrm{N}(q).$$

It identifies $\mathbb{H}_s$ with $M_2(\mathbb{R})$, and its norm-one group with $SL(2,\mathbb{R})$.

Using the Clifford convention $v^2 = Q(v)$, the generators F, G have square $+1$ and anticommute, while $FG = -i$ has square -1. Consequently $\mathbb{H}_s \cong \mathrm{Cl}_{2,0}(\mathbb{R})$. This Clifford realization should not be confused with the quadratic form on the three-dimensional pure subspace: for pure v, $v^2 = -\mathrm{N}(v)I$. See [**51**, Section 7]; notation follows the book conventions.

5. S and $\mathfrak{so}(2,1)$ tensors

The tensor construction of Section 1 identifies the infinitesimal actions of F, G with A_N, Λ_N, while the group lift of $H = F+G$ is Φ_N. These are two different representation constructions, not interchangeable tensor powers.

The symmetric weighted matrix is $S_N = \Phi_N B_N$ (Section 1); the orthogonally normalized involution is U_N of (17). The binomial weights relate these matrices, but do not make their spectral problems identical. In particular, the explicit eigenvectors of Φ_N in Section 5 are not automatically eigenvectors of S_N.

6. Three-dimensional simple Lie algebras

The compact real Lie algebra $\mathfrak{su}(2) \cong \mathfrak{so}(3)$ and the split real algebra $\mathfrak{sl}(2,\mathbb{R}) \cong \mathfrak{so}(1,2)$ are not isomorphic as real Lie algebras. Their complexifications are both $\mathfrak{sl}(2,\mathbb{C})$. In the compact quaternion model all three pure basis units square to -1; in the split model only i does. This is the algebraic distinction between rotations and boosts.

At the group level $SU(2) \to SO(3)$ and $SL(2,\mathbb{R}) \to SO^+(1,2)$ have kernel $\{I, -I\}$. An isomorphism of Lie algebras does not imply an isomorphism of these groups, nor does a statement about the identity component

describe all disconnected orthogonal-group components. See [**51**, Section 7]; notation follows the book conventions.

CHAPTER 8

Lie Groups and Reflections

1. Reflections

Let u be a non-null pure split quaternion. The reflection in the plane Q-orthogonal to u is

$$r_u(v) = -uvu^{-1}.$$

It sends u to $-u$ and fixes any pure v orthogonal to u, since then $uv = -vu$. Null vectors cannot be used in this formula because they are not invertible. Composing two such reflections yields a conjugation action and hence a determinant-one orthogonal map.

For $H = F + G$ one has $H^2 = 2I$, $H^{-1} = H/2$, and

$$HFH^{-1} = G, \quad HGH^{-1} = F, \quad HiH^{-1} = -i.$$

This exchanges the two spacelike axes and reverses the timelike axis. It preserves Q and has determinant one on the pure space, but lies outside the time-orientation-preserving component. See Section 2 for the determinant issue in the spinor space. See [**51**, Section 7]; notation follows the book conventions.

2. Krawtchouk matrices as group elements

Set $H = F + G = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$. Then

$$FH = HG, \qquad GH = HF, \qquad H^2 = 2I.$$

These elementary identities generate the Krawtchouk intertwining and reflection symmetries under symmetric powers.

A qualification to the supplied source is necessary: $\det H = -2$, and $\det(cH) = -2c^2 < 0$ for every nonzero real scalar c. Thus no real scalar normalization puts H in $SL(2,\mathbb{R})$. Likewise $\det F = \det G = -1$. The conjugations are valid in $GL(2,\mathbb{R})$ (or its projective quotient); F, G are also traceless Lie algebra elements. These are distinct statements, and must not be conflated. Over $\mathbb{C}$, $(\mathrm{i}/\sqrt{2})H$ has determinant one, where i denotes the scalar imaginary unit.

The lift to degree-N polynomials is developed once in Section 1; orthogonal normalization is treated in Section 1. Neither construction requires the real matrix H to belong to $SL(2,\mathbb{R})$. See [**51**, Section 7]; notation follows the book conventions.

CHAPTER 9

Representations

1. Splitting formula

For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in GL(2,\mathbb{R})$ define on homogeneous degree-N polynomials

$$(\rho_N(g)p)(x,y) = p(ax+cy, bx+dy).$$

In the basis $e_j = x^{N-j}y^j$, expansion gives its columns, and $\rho_N(gh) = \rho_N(g)\rho_N(h)$. For a Lie algebra element X, instead use the derivative

$$d\rho_N(X) = \frac{d}{dt}\rho_N(I+tX)\Big|_{t=0}.$$

On the full tensor space these two constructions are respectively

$$g \mapsto g^{\otimes N}, \qquad X \mapsto \sum_{r=1}^{N} I^{\otimes(r-1)} \otimes X \otimes I^{\otimes(N-r)}.$$

The group construction is multiplicative; the differential preserves commutators. Tensoring a Lie algebra element multiplicatively would not give the second construction.

For the split units and $H = F + G$, direct differentiation gives

$$d\rho_N(F)e_j = (N-j)e_{j+1} + je_{j-1}, \qquad d\rho_N(G)e_j = (N-2j)e_j.$$

Meanwhile

$$\rho_N(H)e_j = (x+y)^{N-j}(x-y)^j = \sum_{n=0}^{N} K_n(j;N)e_n.$$

Thus $d\rho_N(F) = A_N$, $d\rho_N(G) = \Lambda_N$, and $\rho_N(H) = \Phi_N$. Differentiating the conjugation identity produces $A_N\Phi_N = \Phi_N\Lambda_N$.

The group lift gives a different consequence. Put $J_Ne_j = e_{N-j}$ and $D_Ne_j = (-1)^je_j$. Then

$$\rho_N(F) = J_N, \quad \rho_N(G) = D_N, \quad J_N\Phi_N = \Phi_N D_N, \quad D_N\Phi_N = \Phi_N J_N.$$

Equivalently,

$$K_{N-n}(j;N) = (-1)^jK_n(j;N), \qquad K_n(N-j;N) = (-1)^nK_n(j;N).$$

The master equation and these reflection symmetries therefore arise from the same two-dimensional identity through different lifts. See [51, Section 7]; notation follows the book conventions.

2. Hilbert space structure

With $w_N(j) = 2^{-N}\binom{N}{j}$,

$$\sum_{j=0}^{N} w_N(j) K_n(j;N) K_m(j;N) = \binom{N}{n}\delta_{nm}. \tag{14}$$

Indeed, the weighted product of two generating functions equals $(1+uv)^N$, and coefficient comparison proves (14). Thus

$$\psi_{n,N}(j) = \sqrt{\frac{w_N(j)}{\binom{N}{n}}} K_n(j;N)$$

is an orthonormal basis for the counting-measure space on $\{0,\dots,N\}$.

2.1. Column vectors, covectors, and the binomial metric. For state columns u, v define

$$\langle u, v\rangle_{B^{-1}} = u^T B_N^{-1} v.$$

For row covectors α, β, the dual inner product is $\alpha B_N \beta^T$. The two weights are inverse to one another, not competing choices for the same coordinates. The matrices A_N and Φ_N are self-adjoint for the column metric:

$$A_N^T B_N^{-1} = B_N^{-1} A_N, \qquad \Phi_N^T B_N^{-1} = B_N^{-1}\Phi_N.$$

For A_N this follows directly from the adjacent binomial ratios; for Φ_N it is binomial duality from Section 4. If ϕ_j is column j of Φ_N, then

$$\phi_j^T B_N^{-1} \phi_k = \frac{2^N}{\binom{N}{j}}\delta_{jk}.$$

Thus $2^{-N/2}\sqrt{\binom{N}{j}}\,\phi_j$ are orthonormal in this weighted column space, not generally in ordinary Euclidean coordinates. The coordinate change $u \mapsto B_N^{-1/2}u$ turns this metric into the Euclidean one and yields U_N of (17). See [**51**, Appendix A] for the source's vector/covector viewpoint.

Connection to the supplied article sources

Section 3 ("Krawtchouk matrices and symmetric tensors") of the supplied `KravFeinsKocik.tex`, and the tensor-product and symmetric-tensor appendices of `jkhadamard_Oct_08_00.tex`, supply source material for this chapter. The full tensor space has dimension 2^N, whereas its symmetric sector has dimension $N+1$. Orbit sums, orbit averages, and orthonormal orbit vectors must be distinguished when comparing matrices. See also Chapter 10.

CHAPTER 10

Quantum Probability and Tensor Algebra

1. Flip operator and quantum random walk

On $(\mathbb{R}^2)^{\otimes N}$ let F_r flip the rth binary factor and set $\mathcal{A}_N = \sum_{r=1}^{N} F_r$. In the binary-string basis this is the adjacency operator of the N-cube. Choosing one factor uniformly to flip gives the stochastic operator $\mathcal{A}_N/N$. The symmetric sector is indexed by Hamming weight j and yields the Ehrenfest chain.

Basis normalization matters. For the orbit averages

$$a_j = \binom{N}{j}^{-1} \sum_{|\epsilon|=j} e_\epsilon,$$

one obtains $\mathcal{A}_N a_j = (N-j)a_{j+1} + j a_{j-1}$, hence the book's matrix A_N. The unnormalized orbit sums instead give $(j+1)s_{j+1} + (N-j+1)s_{j-1}$ with endpoint terms omitted. The single flip F is also a real quantum gate, but a unitary quantum walk must be specified separately; the stochastic averaging above is not itself a unitary evolution. See [51, Section 7]; notation follows the book conventions.

2. Krawtchouk matrices as eigenvectors

The local Hadamard identity $FH = HG$ yields

$$\mathcal{A}_N H^{\otimes N} = H^{\otimes N} \mathcal{G}_N, \qquad \mathcal{G}_N = \sum_{r=1}^{N} G_r.$$

A binary string of weight j is an eigenvector of $\mathcal{G}_N$ with eigenvalue $N-2j$. On the orbit-average symmetric sector, this reduces to the master equation proved in Section 2 and interpreted through symmetric powers in Section 1. On the full cube the multiplicity is $\binom{N}{j}$; in the symmetric sector each eigenvalue occurs once. This distinction explains why the $(N+1)$-dimensional Krawtchouk matrix is smaller than the 2^N-dimensional Hadamard tensor matrix. Normalizing H by $1/\sqrt{2}$ gives an orthogonal tensor transform. See [51, Section 7]; notation follows the book conventions.

3. Trace formulas and MacMahon's theorem

TRACE FORMULAS AND MACMAHON'S THEOREM NEEDS TO BE DONE.

4. Chebyshev polynomials

CHEBYSHEV POLYNOMIALS NEEDS TO BE DONE.

5. Supersymmetry

SUPERSYMMETRY NEEDS TO BE DONE.

6. Twistons

The twistons of [51, Section 5, pp. 9–10] are a toy model, not a claim about physical particles. Represent an untwisted band by $\epsilon = +1$ and a Möbius band by $\epsilon = -1$. The model's combination rule multiplies signs: two negative bands give a positive sign, while one negative and one positive give a negative sign. This stipulates parity of twisting; it does not specify a general topological gluing operation.

For N bands, of which j are Möbius bands, define the order-n interaction energy to be the sum of products over all n-element subsets. By (7),

$$E_n = K_n(j; N) = (\Phi_N)_{nj}, \qquad 0 \le n \le N.$$

Column j of Φ_N therefore lists all interaction energies of that configuration. The algebraic identity and its proof belong to Section 1; no second polynomial normalization is introduced here. For $N = 3$ and one Möbius band, the signs $(+1, +1, -1)$ yield

$$(E_0, E_1, E_2, E_3) = (1, 1, -1, -1),$$

which is column 1 of Φ_3 in Appendix A.

Remark 10.1 (Counting convention in the source). Section 5 initially counts nonorientable bands, but its concluding sentence on page 10 says "untwisted bands." Here j consistently counts Möbius bands, as required by (2). Pair interactions mean $\sum_{a<b} \epsilon_a \epsilon_b$, not an unrestricted double sum.

7. Hadamard compression in the Encyclopedia normalization

The supplied `jkhadamard_Oct_08_00.tex`, Sections 4 and 6, and `KravFeinsKocik.tex`, Sections 2–3, connect Hadamard matrices, Krawtchouk matrices, and symmetric tensors. Here is the compression formula in the notation fixed by this book. Set

$$H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \qquad T_N = H^{\otimes N}, \qquad s_j = \sum_{|\epsilon|=j} e_\epsilon, \qquad b_j = \binom{N}{j}, \qquad a_j = s_j / b_j.$$

For an input string of weight j, the sum of the output coefficients at weight n is the coefficient of v^n in $(1+v)^{N-j}(1-v)^j$, namely $K_n(j; N)$. Permutation symmetry then shows that each output coefficient at that weight in $T_N a_j$ is $K_n(j; N)/b_n$. Consequently

$$T_N a_j = \sum_{n=0}^{N} K_n(j; N) a_n.$$

Thus the restriction of T_N to the symmetric sector has matrix Φ_N in the orbit-average basis, not in every choice of basis. In the orthonormal basis $q_j = s_j/\sqrt{b_j}$, the normalized transform $2^{-N/2}T_N$ has matrix

$$2^{-N/2} B_N^{-1/2} \Phi_N B_N^{1/2}.$$

This matrix is orthogonal because $H/\sqrt{2}$ is orthogonal and the symmetric sector is invariant. This is a unitary (real orthogonal) transform, distinct from the stochastic flip average A_N/N. The uploaded articles are preserved as historical source documents; this derivation explicitly fixes the basis and normalization rather than importing their standalone preambles or macros.

CHAPTER 11

Heisenberg Algebra

1. Representations of the Heisenberg algebra

REPRESENTATIONS OF THE HEISENBERG ALGEBRA NEEDS TO BE DONE.

2. Raising and velocity operators; number operator

RAISING AND VELOCITY OPERATORS; NUMBER OPERATOR NEEDS TO BE DONE.

3. Evolution structure and Hamiltonian

EVOLUTION STRUCTURE AND HAMILTONIAN NEEDS TO BE DONE.

4. Time-zero polynomials

TIME-ZERO POLYNOMIALS NEEDS TO BE DONE.

CHAPTER 12

Central Limit Theorem

1. Hermite polynomials

The centered generating function has the exact form

$$G_N(x, v) = (1 - v^2)^{N/2} \exp(x \operatorname{artanh} v).$$

For $x = \sqrt{N}\xi$ and $v = t/\sqrt{N}$, its logarithm tends to $\xi t - t^2/2$. Consequently, for fixed n,

$$N^{-n/2} k_n(\sqrt{N}\xi, N) \longrightarrow \mathrm{He}_n(\xi),$$

where $\exp(\xi t - t^2/2) = \sum_n \mathrm{He}_n(\xi) t^n/n!$. When polynomial values are interpreted only on the original lattice, take admissible sequences approaching ξ.

2. Discrete stochastic differential equations

The exact finite-step precursor of Hermite martingales is (10). In factorial normalization it reads

$$k_n(X_{m+1}, m+1) - k_n(X_m, m) = n\, k_{n-1}(X_m, m)\epsilon_{m+1}.$$

In particular, X_m, $X_m^2 - m$, and $X_m^3 - (3m-2)X_m$ are martingales. The update changes both the spatial variable and the order parameter; it is not the fixed-N urn-chain generator of Chapter 4.

To identify the scaling, define polynomials $H_n(x, t)$ by

$$\exp(ux - tu^2/2) = \sum_{n=0}^{\infty} H_n(x, t) \frac{u^n}{n!}.$$

For $t > 0$, $H_n(x, t) = t^{n/2} \mathrm{He}_n(x/\sqrt{t})$, and coefficient differentiation gives

$$\partial_x H_n = n H_{n-1}, \qquad \partial_t H_n + \tfrac{1}{2}\partial_x^2 H_n = 0.$$

For standard Brownian motion B_t, Ito's formula therefore yields

$$dH_n(B_t, t) = n H_{n-1}(B_t, t)\, dB_t.$$

This is the continuous counterpart of the displayed finite-step equation. The generating-function expansion proves fixed-degree polynomial scaling; a process-level convergence statement additionally requires a functional central limit argument and appropriate integrability control, neither of which is inferred solely from the formal similarity of the equations.

3. Bilinear kernels and the Mehler limit

BILINEAR KERNELS AND THE MEHLER LIMIT NEEDS TO BE DONE.

CHAPTER 13

Clebsch–Gordan Coefficients

1. Clebsch–Gordan coefficients and Krawtchouk polynomials

CLEBSCH–GORDAN COEFFICIENTS AND KRAWTCHOUK POLYNOMIALS NEEDS TO BE DONE.

2. Racah coefficients

RACAH COEFFICIENTS NEEDS TO BE DONE.

3. Clebsch–Gordan coefficients for Heisenberg algebras

CLEBSCH–GORDAN COEFFICIENTS FOR HEISENBERG ALGEBRAS NEEDS TO BE DONE.

CHAPTER 14

Orthogonal Polynomials

1. Three-term recurrence in terms of A, K, and Λ

Differentiating the generating function gives

$$k_{n+1}(x,N) = xk_n(x,N) - n(N-n+1)k_{n-1}(x,N).$$

The spatial equation is

$$\frac{N-x}{2}\big[k_n(x{+}2,N){-}k_n(x,N)\big]{+}\frac{N+x}{2}\big[k_n(x{-}2,N){-}k_n(x,N)\big] = -2nk_n(x,N).$$

These formulas fix the normalization for later operator treatments.

Remark 14.1 (Terminal polynomial and trivial integral zeros). Extend the monic centered recurrence

$$k_{n+1}(x,N) = xk_n(x,N) - n(N-n+1)k_{n-1}(x,N)$$

through $n = N$. The resulting terminal polynomial is

$$k_{N+1}(x,N) = \prod_{j=0}^{N}(x-(N-2j)).$$

Indeed, at each spectral point $x = N{-}2j$, the generating function has degree N, so its coefficient of degree $N+1$ vanishes. The recurrence identifies this coefficient with $k_{N+1}/(N+1)!$, and monicity determines the factorization. Pairing opposite roots gives

$$k_{N+1}(x,N) = \begin{cases} x\displaystyle\prod_{r=1}^{m}(x^2-4r^2), & N=2m, \\ \displaystyle\prod_{r=0}^{m}(x^2-(2r+1)^2), & N=2m+1. \end{cases}$$

Empty products are one. These are the trivial integral zeros: the entire finite spectrum $-N, -N{+}2, \ldots, N$, with spacing two. In ordinary-coefficient normalization the corresponding centered polynomial is $k_{N+1}(x,N)/(N+1)!$. The terminal polynomial vanishes on the support and has zero squared norm; it is not an additional positive-norm member of the orthogonal system.

1.1. Matrix form of the two recurrences. With A_N, R_N, L_N, Λ_N defined in Section 1, and $\Phi_N = (K_n(j;N))$ as in Section 3.1, the degree and spatial recurrences take the paired form

$$A_N\Phi_N = \Phi_N\Lambda_N, \qquad \Phi_N A_N = \Lambda_N\Phi_N. \tag{15}$$

The coefficient proof and entrywise recurrences are given in Section 2; the tensor interpretation is in Section 1. Terms outside the degree or column range are omitted. Thus the columns of Φ_N are right eigenvectors of A_N, and its rows are left eigenvectors, with the equally spaced spectrum $N, N-2, \dots, -N$. This does not make A_N symmetric in the standard basis. Explicit examples appear in Appendix A.

2. Nonsymmetric case

To avoid silently changing the symmetric normalization, introduce a new family. Let $0 < p < 1$, $q = 1 - p$, and define

$$(1+pv)^{N-j}(1-qv)^j = \sum_{n=0}^{N} Q_n^{(p)}(j;N)v^n. \tag{16}$$

If $J \sim \mathrm{Bin}(N,p)$, a single factor takes the value $1+pv$ with probability q and $1-qv$ with probability p. Its linear coefficient is centered and has variance pq. Therefore

$$\mathbb{E}[Q_n^{(p)}(J;N)Q_m^{(p)}(J;N)] = \delta_{nm}\binom{N}{n}(pq)^n.$$

This is the two-point instance of Section 3. At $p = q = 1/2$, $Q_n^{(1/2)} = 2^{-n}K_n$; the new family is not numerically identical to K_n. Writing $a = pN - j$, logarithmic differentiation of (16) gives

$$(n+1)Q_{n+1}^{(p)} = [a - (p-q)n]Q_n^{(p)} - pq(N-n+1)Q_{n-1}^{(p)}.$$

For example, $Q_0^{(p)} = 1$, $Q_1^{(p)} = a$, and $2Q_2^{(p)} = a^2 - (p-q)a - pqN$.

The reversible birth–death generator on $0 \le j \le N$ is

$$(\mathcal{L}_p f)(j) = p(N-j)[f(j+1) - f(j)] + qj[f(j-1) - f(j)].$$

Boundary terms with zero rates are omitted. Binomial detailed balance follows from $\pi_p(j)p(N-j) = \pi_p(j+1)q(j+1)$, where $\pi_p(j) = \binom{N}{j}p^j q^{N-j}$. Acting on the generating function gives $\mathcal{L}_p G = -v\partial_v G$, hence

$$\mathcal{L}_p Q_n^{(p)} = -nQ_n^{(p)}.$$

This generator uses a one-coordinate refresh rate convention. At $p = 1/2$ it is half the sign-flip generator whose eigenvalues are $-2n$ in Section 1. The factor is a time normalization, not a contradiction.

CHAPTER 15

Krawtchouk Transforms

1. Orthogonal transformation associated to K

The raw Krawtchouk matrix Φ_N is not generally orthogonal in the Euclidean inner product. Its orthogonal normalization is

$$U_N = 2^{-N/2} B_N^{-1/2} \Phi_N B_N^{1/2}, \qquad B_N = \operatorname{diag}\binom{N}{j}. \tag{17}$$

Indeed, binomial duality and weighted orthogonality give

$$\Phi_N B_N = B_N \Phi_N^T, \qquad \Phi_N B_N \Phi_N^T = 2^N B_N.$$

Consequently $U_N^T = U_N$ and $U_N^T U_N = I$. This distinguishes the raw matrix Φ_N, its orthogonal normalization U_N, and the symmetric weighted matrix S_N of Section 1. In particular,

$$S_N = 2^{N/2} B_N^{1/2} U_N B_N^{1/2},$$

is a congruence, not a similarity.

2. Exponential function in the Krawtchouk basis

Let $x_j = N - 2j$ and $t \in \mathbb{R}$. Since $e^{t\epsilon} = \cosh t\,(1 + \epsilon \tanh t)$ for $\epsilon \in \{+1, -1\}$, multiplication over N signs gives the exact finite expansion

$$e^{t x_j} = (\cosh t)^N \sum_{n=0}^{N} (\tanh t)^n K_n(j; N). \tag{18}$$

For complex t a denominator-free version is preferable:

$$e^{t x_j} = \sum_{n=0}^{N} (\sinh t)^n (\cosh t)^{N-n} K_n(j; N).$$

It holds even when $\cosh t = 0$. Orthogonality from Section 2 then gives

$$\mathbb{E}[e^{t X_N} K_n(J; N)] = \binom{N}{n} (\sinh t)^n (\cosh t)^{N-n}.$$

For $N = 2$ this reads $e^{tx} = \cosh^2 t + x \sinh t \cosh t + (x^2 - 2)\sinh^2 t/2$ on $x = 2, 0, -2$. These are expansions of lattice functions; they do not identify the raw transform Φ_N with the Euclidean-orthogonal normalization U_N.

3. Krawtchouk transform

3.1. The ordinary-generating-function matrix. We use the row index n for degree and the column index j for the lattice argument:

$$(1+v)^{N-j}(1-v)^j = \sum_{n=0}^{N} K_n(j;N)v^n, \qquad \Phi_N = (K_n(j;N))_{n,j=0}^{N}. \tag{19}$$

For example,

$$\Phi_2 = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}.$$

The transform is an involution up to scale:

$$\Phi_N^2 = 2^N I, \qquad \Phi_N^{-1} = 2^{-N}\Phi_N. \tag{20}$$

For completeness, summing a fixed row of the matrix square via the generating function gives

$$\begin{aligned} \sum_{n=0}^{N} v^n(\Phi_N^2)_{nj} &= \sum_{k=0}^{N}(1+v)^{N-k}(1-v)^k K_k(j;N) \\ &= (1+v)^N \left(1+\frac{1-v}{1+v}\right)^{N-j}\left(1-\frac{1-v}{1+v}\right)^{j} = 2^N v^j. \end{aligned}$$

The apparent denominator at $v=-1$ is removed by the polynomial identity. Comparing coefficients proves (20). The recurrence identities (15) connect this transform to the finite Lie-algebra generators. Tables for $N = 2,3,4,5,7,8$ are in Appendix A.

4. Krawtchouk transform and FFT

KRAWTCHOUK TRANSFORM AND FFT NEEDS TO BE DONE.

5. Binomial bases and dual transforms

Let $P_N = (\binom{k}{i})_{i,k=0}^{N}$ and denote column k by $c^{(k)}$:

$$c_i^{(k)} = \binom{k}{i}, \qquad c_i^{(k)} = 0 \quad (i>k).$$

The Pascal matrix P_N is triangular with diagonal entries one and must not be confused with the diagonal binomial matrix B_N. For $f_a(v) = \sum_{j=0}^{N} a_j v^j$, the defining transform gives

$$f_{\Phi_N a}(v) = (1+v)^N f_a\left(\frac{1-v}{1+v}\right).$$

Apply this to $f_{c^{(k)}}(v) = (1+v)^k$ to obtain

$$\Phi_N c^{(k)} = 2^k c^{(N-k)}. \tag{21}$$

The apparent denominator is removable because both sides are polynomial. Writing $J_N e_k = e_{N-k}$ and $W_N = \mathrm{diag}(2^0, \dots, 2^N)$, this is

$$P_N^{-1} \Phi_N P_N = J_N W_N.$$

Here W_N is used to avoid collision with the parity matrix D_N of Section 1. This is the existing binomial-basis identity, with its proof made explicit; compare [51, Section 7.3].

5.1. Explicit eigenvectors of the raw transform. For each $0 \le k < N/2$, define

$$v_k^{\pm} = 2^{(N-k)/2} c^{(k)} \ \pm \ 2^{k/2} c^{(N-k)}.$$

Equation (21) gives

$$\Phi_N v_k^{\pm} = \pm 2^{N/2} v_k^{\pm}.$$

Use each complementary pair only once. If N is even, the unpaired column $c^{(N/2)}$ has eigenvalue $+2^{N/2}$; the corresponding minus combination would be zero and must not be counted. Since the columns of P_N are independent, these vectors form a complete basis. The multiplicities are

$$m_+ = \lfloor N/2 \rfloor + 1, \qquad m_- = \lfloor (N+1)/2 \rfloor.$$

For $N = 3$ both multiplicities are two; for $N = 5$ both are three.

These are eigenvectors of Φ_N, not the columns of Φ_N viewed as eigenvectors of A_N, and not eigenvectors of S_N in general. For the orthogonal involution U_N, the corresponding eigenvectors are $B_N^{-1/2} v_k^{\pm}$, with eigenvalues ± 1. Pascal coordinates are not orthonormal: Euclidean inner products in these coordinates use the Gram matrix $P_N^T P_N$.

6. Kravchuk sums

This section collects sums of entries, squares, and absolute values of the raw matrix $\Phi_N = (K_n(j; N))$. The attached estimate note is preserved unchanged as `sources/KravEstimate.tex`. Selected exact formulas also occur in the retained local sums manuscripts in `supporting-papers/`; the proofs below fix the book's normalization and do not import those standalone documents' preambles or bibliography keys. These formulas concern Φ_N, not its symmetric weighted form $S_N = \Phi_N B_N$ or orthogonal normalization U_N.

6.1. A dictionary of sums. Write $b_j = \binom{N}{j}$ and define

$$R_n = \sum_{j=0}^{N} K_n(j;N), \qquad C_j = \sum_{n=0}^{N} K_n(j;N),$$
$$Q_n^{\mathrm{row}} = \sum_{j=0}^{N} K_n(j;N)^2, \qquad Q_j^{\mathrm{col}} = \sum_{n=0}^{N} K_n(j;N)^2,$$
$$A_n^{\mathrm{row}} = \sum_{j=0}^{N} |K_n(j;N)|, \qquad A_j^{\mathrm{col}} = \sum_{n=0}^{N} |K_n(j;N)|.$$

The corresponding whole-matrix totals are $T_1(N) = \sum_{n,j} K_n(j;N)$, $T_2(N) = \sum_{n,j} K_n(j;N)^2$, and $T_{\mathrm{abs}}(N) = \sum_{n,j} |K_n(j;N)|$. Unless explicitly stated, these are unweighted sums. They must not be identified with binomially weighted expectations.

6.2. Signed sums and discrete antiderivatives. The complete signed sums are

(22)
$$C_j = \begin{cases} 2^N, & j = 0, \\ 0, & 1 \le j \le N, \end{cases} \qquad R_n = \begin{cases} \binom{N+1}{n+1}, & n \text{ even}, \\ 0, & n \text{ odd}, \end{cases} \qquad T_1(N) = 2^N.$$

Evaluate the column generating function at $v = 1$ to obtain C_j. For the row sum, the finite geometric identity gives

$$\sum_{j=0}^{N} (1+v)^{N-j}(1-v)^j = \frac{(1+v)^{N+1} - (1-v)^{N+1}}{2v};$$

the singularity at zero is removable, and coefficient comparison gives R_n. For $1 \le j \le N$ and $0 \le m \le N$, division by $1 - v$ gives the partial column sum

$$\sum_{n=0}^{m} K_n(j;N) = K_m(j-1;N-1). \tag{23}$$

Here coefficients of degree above $N - 1$ vanish. The cross-order identities also give the spatial antiderivative

$$K_{n+1}(j;N+1) - K_{n+1}(j+1;N+1) = 2K_n(j;N). \tag{24}$$

Thus truncated signed row sums telescope, and the same formula evaluates absolute row sums interval by interval once the signs are known.

6.3. Weighted squares versus raw square sums. The attachment correctly derives from binomial duality and orthogonality that

$$\sum_{j=0}^{N} b_j K_n(j;N)^2 = 2^N b_n, \qquad \sum_{n=0}^{N} \frac{K_n(j;N)^2}{b_n} = \frac{2^N}{b_j}. \tag{25}$$

These are weighted identities, not formulas for the raw row or column square sums. For the latter one has

$$Q_j^{\mathrm{col}} = \frac{\binom{2j}{j}\binom{2N-2j}{N-j}}{\binom{N}{j}} = \frac{(2j)!(2N-2j)!}{j!(N-j)!N!}, \tag{26}$$

and

$$Q_n^{\mathrm{row}} = \sum_{k=0}^{\min(n,\lfloor N/2\rfloor)} \binom{N+1}{2k+1}\binom{2k}{k}\binom{N-2k}{n-k}. \tag{27}$$

Consequently,

$$T_2(N) = \sum_{j=0}^{N} \frac{\binom{2j}{j}\binom{2N-2j}{N-j}}{\binom{N}{j}} = \sum_{k=0}^{\lfloor N/2\rfloor} \binom{N+1}{2k+1}\binom{2k}{k} 2^{N-2k}. \tag{28}$$

For (26), apply coefficient Parseval to $(1+v)^{N-j}(1-v)^j$ on the unit circle. Substitution $v = e^{i\theta}$ reduces the integral to $4^N \mathrm{B}(j+1/2, N-j+1/2)/\pi$, whose half-integer gamma factors give the displayed quotient. For (27), put $A = (1+u)(1+v)$ and $B = (1-u)(1-v)$. Then

$$\sum_{j=0}^{N} G_{N,j}(u)G_{N,j}(v) = \frac{A^{N+1}-B^{N+1}}{A-B} = \sum_{k=0}^{\lfloor N/2\rfloor} \binom{N+1}{2k+1}(1+uv)^{N-2k}(u+v)^{2k}.$$

Extract $[u^n v^n]$ and then sum over n for (28). Examples are $Q_0^{\mathrm{row}} = N+1$ and $Q_1^{\mathrm{row}} = N(N+1)(N+2)/3$.

6.4. Absolute sums: bounds and exact boundary columns. Cauchy–Schwarz applied to (25) gives the attachment's valid estimates

$$A_n^{\mathrm{row}} \le \left(2^N b_n \sum_{j=0}^{N} b_j^{-1}\right)^{1/2}, \qquad A_j^{\mathrm{col}} \le \frac{2^N}{\sqrt{b_j}}. \tag{29}$$

For $N \ge 1$ the endpoint columns each contribute 2^N, so the correct summed bound is

$$2^{N+1} \le T_{\mathrm{abs}}(N) \le 2^N \sum_{j=0}^{N} \binom{N}{j}^{-1/2} = 2^N \left(2 + \sum_{j=1}^{N-1} \binom{N}{j}^{-1/2}\right). \tag{30}$$

The summation on the upper bound is essential. For $N \ge 6$, unimodality gives the explicit estimate

$$\sum_{j=1}^{N-1} \binom{N}{j}^{-1/2} \le \frac{2}{\sqrt{N}} + \frac{2}{\sqrt{\binom{N}{2}}} + \frac{N-5}{\sqrt{\binom{N}{3}}} = O(N^{-1/2}).$$

Thus $T_{\mathrm{abs}}(N) \sim 2^{N+1}$, as in the attachment. For $N = 0$ the single-entry total is one, so the two-endpoint lower bound is not applicable.

More precisely,

$$A_0^{\mathrm{col}} = A_N^{\mathrm{col}} = 2^N, \qquad A_1^{\mathrm{col}} = A_{N-1}^{\mathrm{col}} = 2\binom{N-1}{\lfloor (N-1)/2 \rfloor} \quad (N \geq 1). \tag{31}$$

Indeed, $K_n(1;N) = \binom{N-1}{n} - \binom{N-1}{n-1}$, and the absolute first differences of a unimodal sequence, extended by zero at both ends, sum to twice its maximum. For even N, this equals $\binom{N}{N/2}$. The attachment's general near-boundary-column expression misses the factor two. When $N \geq 3$ the two near-boundary columns are distinct and give the corrected improvement

$$T_{\mathrm{abs}}(N) \geq 2^{N+1} + 4\binom{N-1}{\lfloor (N-1)/2 \rfloor}. \tag{32}$$

At $N = 2$ there is just one middle column and it must not be counted twice. The next boundary column has, for $N > 2$,

$$A_2^{\mathrm{col}} = A_{N-2}^{\mathrm{col}} = 4\left[\binom{N-2}{r_N} - \binom{N-2}{r_N - 1}\right], \qquad r_N = \left\lfloor \frac{N - \sqrt{N}}{2} \right\rfloor. \tag{33}$$

To see this, use

$$K_n(2;N) = \binom{N}{n}\frac{(N-2n)^2 - N}{N(N-1)} = \binom{N-2}{n} - 2\binom{N-2}{n-1} + \binom{N-2}{n-2}.$$

The first expression locates the two positive tails, the second telescopes on a tail, and the zero signed column total makes the full absolute sum four times the left-tail mass. Zero entries at an integral root cause no ambiguity. For even $N = 2m$, the middle column additionally has $A_m^{\mathrm{col}} = 2^m$, directly from $(1-v^2)^m$. These are entrywise absolute sums. In contrast, the induced matrix norm is the maximum column absolute sum, $\|\Phi_N\|_1 = 2^N$.

For a row of constant sign ϵ on $a \leq j \leq b$, (24) gives

$$\sum_{j=a}^{b} |K_n(j;N)| = \frac{\epsilon}{2}\big[K_{n+1}(a;N+1) - K_{n+1}(b+1;N+1)\big].$$

Summing over constant-sign intervals is an exact finite evaluation of A_n^{row}, without replacing it by a weighted expectation.

6.5. Boundary expansion of the total absolute sum. The selected sums manuscript sharpens the attachment's leading equivalence to

$$T_{\mathrm{abs}}(N) = 2^{N+1}\left[1 + \sqrt{\frac{2}{\pi N}} + \frac{4}{\sqrt{2\pi e}\, N} + O(N^{-3/2})\right]. \tag{34}$$

Here is the mechanism. For any fixed positive integer q,

$$\sum_{j=q}^{N-q} \binom{N}{j}^{-1/2} = O(N^{-q/2}).$$

On the range $q \leq j \leq N/4$, the ratio of consecutive terms is $\sqrt{(j+1)/(N-j)}$, bounded away from one for large N, so the sum is controlled by its first term.

Between $N/4$ and $N/2$ the terms are exponentially small; reflection treats the other half. Hence the columns $3 \le j \le N-3$ contribute $O(2^N N^{-3/2})$. Separating the first three columns at each end gives

$$\frac{T_{\mathrm{abs}}(N)}{2^{N+1}} = 1 + \frac{A_1^{\mathrm{col}}}{2^N} + \frac{A_2^{\mathrm{col}}}{2^N} + O(N^{-3/2}).$$

Stirling's formula in (31) gives $A_1^{\mathrm{col}}/2^N = \sqrt{2/(\pi N)} + O(N^{-3/2})$. For (33), write the binomial difference as

$$\binom{N-2}{r_N} \frac{N-1-2r_N}{N-1-r_N}.$$

Since $r_N = N/2 - \sqrt{N}/2 + O(1)$, the two factors are respectively $2^{N-2}\sqrt{2/(\pi N)}e^{-1/2}(1 + O(N^{-1/2}))$ and $2N^{-1/2}(1 + O(N^{-1/2}))$. Including the factor four proves the N^{-1} term. This is a boundary-column expansion, not an asymptotic interchange of an uncontrolled full matrix sum.

6.6. Exact numerical checks. The following totals are computed with integer arithmetic; T_2 is a raw, unweighted sum. The companion verification record checks signed sums, raw and weighted squares, absolute column bounds and boundary formulas through $N = 32$.

N	$T_1(N)$	$T_2(N)$	$T_{\mathrm{abs}}(N)$
5	32	584	104
10	1024	394404	3016
15	32768	322494208	88656
20	1048576	283443548276	2655112
32	4294967296	3726566597989349958	10181521568

For the unchanged attachment, corrections and integration scope, see `sources/INTEGRATION_KravEsti`
The local supporting manuscripts remain preserved, not wholesale certified by their inclusion in the project.

CHAPTER 16

Hypergeometric Functions

1. Krawtchouk polynomials as hypergeometric functions

KRAWTCHOUK POLYNOMIALS AS HYPERGEOMETRIC FUNCTIONS NEEDS TO BE DONE.

2. Addition formulas

ADDITION FORMULAS NEEDS TO BE DONE.

CHAPTER 17

Symmetric Krawtchouk Matrices

1. Definition and small-order examples

With the row/column convention of (19), put

$$S_N = \Phi_N B_N, \qquad (S_N)_{nj} = \binom{N}{j} K_n(j;N). \tag{35}$$

Thus S_N is obtained by multiplying column j of Φ_N by $\binom{N}{j}$, not by multiplying its row j. Binomial duality gives

$$\binom{N}{j} K_n(j;N) = \binom{N}{n} K_j(n;N), \qquad S_N^T = S_N.$$

For $N = 2$,

$$S_2 = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ 1 & -2 & 1 \end{pmatrix}.$$

For $N = 4$, the column weights are $(1, 4, 6, 4, 1)$ and

$$S_4 = \begin{pmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 8 & 0 & -8 & -4 \\ 6 & 0 & -12 & 0 & 6 \\ 4 & -8 & 0 & 8 & -4 \\ 1 & -4 & 6 & -4 & 1 \end{pmatrix}.$$

Although $\Phi_N^2 = 2^N I$, the inserted binomial weights prevent identifying S_N^2 with $2^N I$. Nor is S_N the same matrix as the orthogonal normalization U_N of (17). Complete tables of $A_N, R_N, L_N, \Phi_N, S_N$ are in Appendix A.

2. The matrix T

THE MATRIX T NEEDS TO BE DONE.

3. S^2 and trace formulas

Binomial duality yields the exact bilinear identity

$$(S_N^2)_{jk} = \binom{N}{j}\binom{N}{k} \sum_{n=0}^{N} K_n(j;N) K_n(k;N).$$

Parseval applied to the generating functions evaluates it as

(36)
$$(S_N^2)_{jk} = \frac{4^N}{\pi}\binom{N}{j}\binom{N}{k}\cos\frac{\pi(j-k)}{2}\,\mathrm{B}\left(\frac{j+k+1}{2},\frac{2N-j-k+1}{2}\right).$$

The beta integral follows by factoring $1 \pm e^{i\theta}$ into half-angle sine and cosine factors. This identity is exact, not an asymptotic statement.

4. Spectrum of S

SPECTRUM OF S NEEDS TO BE DONE.

5. Symmetry sectors and block decompositions

Define $(Jf)_j = f_{N-j}$ and $(Ef)_j = (-1)^j f_j$. Then

$$JS_N = S_N E, \qquad S_N J = E S_N,$$

so S_N commutes with $J + E$. For the square, indices of different parity decouple. Conjugation by $\mathrm{diag}((-1)^{\lfloor j/2 \rfloor})$ makes each parity block entrywise positive, by (36).

6. Beta-function kernel and spectral asymptotics

Remark 17.1 (Working asymptotic program, not a completed theorem). Put $\xi = (N-2j)/\sqrt{N}$ and $\eta = (N-2k)/\sqrt{N}$. Compact-window Stirling expansions of (36), after sign removal and normalization by $N/8^N$, lead to the candidate kernel

$$L(\xi,\eta) = \frac{1}{\sqrt{2}\pi^{3/2}}\exp\left[-\frac{3}{8}(\xi^2+\eta^2)+\frac{1}{4}\xi\eta\right].$$

Its Gaussian ground state is $\exp(-\xi^2/(2\sqrt{2}))$, with eigenvalue $\kappa = (\sqrt{8}-2)/\pi$. The proposed positive-eigenvalue asymptotic is

$$\lambda_{\max}(S_N) \sim \sqrt{\sqrt{8}-2}\,\frac{(\sqrt{8})^N}{\sqrt{\pi N}}.$$

Global operator convergence and control of the positive spectral branch still require proof. Fixed-degree Hermite limits alone do not justify this conclusion.

CHAPTER 18

Gaussian Quadrature

Source and derivation note. The orthogonal system and normalization are those of Sections 2, 2, and 1. The product-martingale source is the supplied `sources/KravFeinsKocik.tex`, identified in the source register. The Gaussian rule, terminal factorization, proofs, and numerical examples here are developed from those identities; they are not attributed to an unread item in the supplied literature list. Numerical provenance and reproduction details are in `sources/INTEGRATION_gaussian_summation.md` and `sources/compute_gaussian_krawtchouk.py`.

1. Zeros of Krawtchouk polynomials

The measure in this chapter is finite and discrete:

$$\mu_N = 2^{-N} \sum_{j=0}^{N} \binom{N}{j} \delta_j, \qquad \mathcal{S}_N[f] = 2^{-N} \sum_{j=0}^{N} \binom{N}{j} f(j).$$

Thus Gaussian quadrature here compresses a weighted sum, not an ordinary integral. Keep two integers distinct: N specifies the $N+1$ support points, whereas n specifies the degree of the orthogonal polynomial and the number of Gaussian nodes. We take $1 \le n \le N$ unless stated otherwise.

Extend $K_n(j; N)$ polynomially off the integer lattice via the recurrence of Section 3. Its leading coefficient in j is $(-2)^n/n!$. The monic polynomials

$$p_n(j) = \frac{(-1)^n n!}{2^n} K_n(j; N) = \frac{(-1)^n}{2^n} k_n(N - 2j, N)$$

therefore satisfy

$$p_{r+1}(j) = (j - N/2)p_r(j) - \frac{r(N - r + 1)}{4} p_{r-1}(j), \qquad p_0 = 1. \tag{37}$$

Their squared norms under μ_N are

$$h_r = \mathcal{S}_N[p_r^2] = \frac{(r!)^2}{4^r} \binom{N}{r} = \frac{r!N!}{4^r (N - r)!}. \tag{38}$$

The Gaussian nodes $z_1 < \cdots < z_n$ are the zeros of p_n, equivalently the eigenvalues of the $n \times n$ real symmetric Jacobi matrix

(39)
$$\mathcal{J}_{N,n} = \begin{pmatrix} N/2 & b_1 & & & \\ b_1 & N/2 & b_2 & & \\ & b_2 & N/2 & \ddots & \\ & & \ddots & \ddots & b_{n-1} \\ & & & b_{n-1} & N/2 \end{pmatrix}, \qquad b_r = \frac{1}{2}\sqrt{r(N-r+1)}.$$

For $n = 1$ this means the scalar matrix $(N/2)$. Positivity of b_r gives simple real eigenvalues; the Rayleigh quotient is the weighted average of j against the square of a nonzero polynomial of degree at most $n-1$. Such a polynomial cannot vanish on all N other support points, so $0 < z_a < N$. The zeros of successive degrees strictly interlace. Reflection symmetry gives $z_{n+1-a} = N - z_a$. These nodes are generally not integers: polynomial extension, rather than a table of lattice values alone, is essential.

2. Integral zeros

The trivial terminal zeros. Remark 14.1 gives a universal family of integral zeros at the terminal degree:

$$k_{N+1}(x, N) = \prod_{j=0}^{N} (x - (N - 2j)).$$

For even $N = 2m$ this is $x\prod_{r=1}^{m}(x^2 - 4r^2)$; for odd $N = 2m+1$ it is $\prod_{r=0}^{m}(x^2 - (2r+1)^2)$. Thus all centered spectral points are zeros. For example,

$$k_6(x, 5) = (x^2 - 1)(x^2 - 9)(x^2 - 25).$$

In the uncentered coordinate $j = (N - x)/2$, this is precisely the monic terminal polynomial $p_{N+1}(j) = \prod_{r=0}^{N}(j - r)$ discussed below. This terminal case is automatic and should be distinguished from the nontrivial question of integral zeros in degrees $1 \le n \le N$.

Integer zeros are special cases of these polynomial zeros, not a prerequisite for Gaussian summation. Since $K_n(N - j; N) = (-1)^n K_n(j; N)$, every odd degree has the central zero $j = N/2$; this is integral exactly when N is even. For degree two,

$$K_2(j; N) = \frac{(N - 2j)^2 - N}{2}, \qquad z_{\pm} = \frac{N \pm \sqrt{N}}{2}.$$

For $N \ge 2$ these two zeros are integers exactly when N is a perfect square: a square and its square root have the same parity. For example, $N = 9$ gives nodes 3 and 6, each with Gaussian weight $1/2$. Consequently the two-node rule integrates, in the discrete sense, every cubic exactly under μ_9. For $N = 5$ the degree-three centered polynomial is $k_3(x, 5) = x(x^2 - 13)$, giving

$j = (5-\sqrt{13})/2, 5/2, (5+\sqrt{13})/2$, none integral. These examples are not a classification of integral zeros at higher degree.

A GENERAL CLASSIFICATION OF INTEGRAL ZEROS NEEDS TO BE DONE.

3. Gaussian–Krawtchouk summation

For each Gaussian node define the Lagrange polynomial

$$\ell_a(j) = \frac{p_n(j)}{(j-z_a)p_n'(z_a)}, \qquad \lambda_a = \mathcal{S}_N[\ell_a].$$

Then the n-node formula is

$$2^{-N}\sum_{j=0}^{N}\binom{N}{j}f(j) = \sum_{a=1}^{n}\lambda_a f(z_a), \qquad \deg f \le 2n-1. \tag{40}$$

To prove exactness, divide $f = qp_n + r$ with $\deg q, \deg r \le n-1$. Orthogonality annihilates $\mathcal{S}_N[qp_n]$, while interpolation of r evaluates its sum exactly. Since $p_n(z_a) = 0$, the right side is also unchanged by removing qp_n. Applying exactness to ℓ_a^2 shows $\lambda_a = \mathcal{S}_N[\ell_a^2] > 0$, and applying it to 1 gives $\sum_a \lambda_a = 1$. The rule is symmetric: $\lambda_{n+1-a} = \lambda_a$. Two equivalent computable weight formulas are

$$\lambda_a = \left[\sum_{r=0}^{n-1}\frac{K_r(z_a;N)^2}{\binom{N}{r}}\right]^{-1} = \frac{h_{n-1}}{p_n'(z_a)p_{n-1}(z_a)}. \tag{41}$$

Alternatively, if u_a is the unit eigenvector of (39) for z_a, then $\lambda_a = |(u_a)_0|^2$. The spectral measure of the first coordinate gives this identity; the Christoffel formula follows by summing squares of orthonormal polynomials. These formulas use the probability-normalized binomial measure. For the unnormalized sum $\sum_j \binom{N}{j}f(j)$ multiply every weight by 2^N.

3.1. Error and the finite-support endpoint. For $f \in C^{2n}([0,N])$, Hermite interpolation at the n nodes, followed by summation against the positive measure, gives

$$\mathcal{S}_N[f] - \sum_a \lambda_a f(z_a) = \frac{h_n}{(2n)!}f^{(2n)}(\xi), \qquad \xi \in [0,N].$$

In particular, the error on the monic polynomial j^{2n} is exactly $h_n > 0$, so degree $2n-1$ is sharp. The rule remains meaningful for nonpolynomial functions provided values between lattice points are defined. If only the $N+1$ discrete data values are known, choosing an off-lattice interpolation is extra information; one cannot assume an arbitrary data array has prescribed Gaussian-node values. For $n = N+1$, the terminal polynomial is $p_{N+1}(j) = \prod_{r=0}^{N}(j-r)$ and its norm is zero. The full Jacobi matrix recovers the original nodes $0,\dots,N$ and weights $2^{-N}\binom{N}{j}$. This is the original finite sum, exact for

every lattice function; it is not a new compressed rule, and the positive-norm error discussion above is restricted to $n \le N$.

3.2. Accurate examples for $N = 5, 10, 15$. The following tables give both reduced-node rules $(N, n) = (5, 3), (10, 5), (15, 8)$ and the degree-N rules $(5, 5), (10, 10), (15, 15)$. Thus the reduced rules replace respectively $6, 11, 16$ summands by $3, 5, 8$ evaluations and are exact through degrees $5, 9, 15$. The degree-N rules are exact through degrees $9, 19, 29$. All nodes below are in the original j coordinate; centered nodes are $x_a = N - 2z_a$ with the same weights.

For $(N, n) = (5, 3)$ one also has the exact formula

$$\mathcal{S}_5[f] = \frac{5}{26} f\left(\frac{5 - \sqrt{13}}{2}\right) + \frac{8}{13} f\left(\frac{5}{2}\right) + \frac{5}{26} f\left(\frac{5 + \sqrt{13}}{2}\right), \qquad \deg f \le 5.$$

The tables show 18 significant digits. Their machine-readable companions retain 65 digits. Computation used a symmetric Jacobi eigensystem at 90-decimal working precision, independently checked against the centered polynomial recurrence, the Christoffel weight formula, positivity, and every scaled moment $(j/N)^r$ for $0 \le r \le 2n - 1$. Maximum absolute scaled-moment error over all six rules was below 10^{-85} at working precision. These numerical residuals are consistency checks, not interval-certified error bounds; the printed rounded tables have correspondingly lower accuracy.

$N = 5$, $n = 3$; **exact through degree** 5**.**

a	z_a	λ_a
1	0.697224362268005327	0.192307692307692318
2	2.5	0.615384615384615419
3	4.30277563773199478	0.192307692307692318

$N = 5$, $n = 5$; **exact through degree** 9**.**

a	z_a	λ_a
1	0.0649539076702599777	0.0406121317521831102
2	1.24678392596102428	0.244622767576675948
3	2.5	0.42953020134228187
4	3.7532160740389755	0.244622767576675948
5	4.93504609232974012	0.0406121317521831102

$N = 10$, $n = 5$; **exact through degree** 9**.**

a	z_a	λ_a
1	1.0	0.01953125
2	3.0	0.234375
3	5.0	0.4921875
4	7.0	0.234375
5	9.0	0.01953125

$N = 10$, $n = 10$; **exact through degree** 19**.**

a	z_a	λ_a
1	0.00426140162761967471	0.00100005627585995687
2	1.03215639227186573	0.0108643711646248114
3	2.1094239579268903	0.0553578832768971893
4	3.24085911987510711	0.160428693955712715
5	4.40940159989234548	0.272348995326905319
6	5.59059840010765452	0.272348995326905319
7	6.75914088012489334	0.160428693955712715
8	7.8905760420731097	0.0553578832768971893
9	8.96784360772813471	0.0108643711646248114
10	9.99573859837238032	0.00100005627585995687

$N = 15$, $n = 8$; **exact through degree** 15**.**

a	z_a	λ_a
1	0.810168654648509734	0.000551841058012324075
2	2.70329267453015554	0.0178684857525261559
3	4.61682155336053412	0.134507504747883117
4	6.53821132811295858	0.347072168441578399
5	8.46178867188704054	0.347072168441578399
6	10.3831784466394659	0.134507504747883117
7	12.2967073254698445	0.0178684857525261559
8	14.1898313453514895	0.000551841058012324075

$N = 15$, $n = 15$; **exact through degree** 29**.**

a	z_a	λ_a
1	0.00021067529406323948 2	0.000030559226494242770 8
2	1.00264068454641486	0.000462988590523494585
3	2.01486605080593995	0.00335579980348639896
4	3.05052871756057487	0.0155659622211499205
5	4.12061368475752943	0.0506438220470849867
6	5.22535900511131501	0.117850851005629698
7	6.35599416357050373	0.195969221661716791
8	7.5	0.232241590887828919
9	8.64400583642949627	0.195969221661716791
10	9.77464099488868499	0.117850851005629698
11	10.8793863152424706	0.0506438220470849867
12	11.9494712824394256	0.0155659622211499205
13	12.9851339491940596	0.00335579980348639896
14	13.9973593154535845	0.000462988590523494585
15	14.9997893247059366	0.000030559226494242770 8

CHAPTER 19

Coding Theory

1. MacWilliams's theorem

MACWILLIAMS'S THEOREM NEEDS TO BE DONE.

2. Association schemes

ASSOCIATION SCHEMES NEEDS TO BE DONE.

CHAPTER 20

Source conventions and the general Krawtchouk transform

This chapter integrates the supplied summary paper *On Krawtchouk Transforms* (`KravPQ.tex`) with `KravSums.tex`. The originals are retained in `sources/`. Source conventions are preferred here; explicit dictionaries keep the earlier symmetric chapters compatible. Statements below are derived directly from the supplied generating functions, rather than imported with an implicit normalization.

1. The two source conventions

In the transform paper, let $p, q > 0$, $p + q = 1$, and $x = N - 2j$. Its convention is

$$G(v) = (1 + 2qv)^{N-j}(1 - 2pv)^j = \sum_{n=0}^{N} \frac{v^n}{n!} K_n(x, N). \tag{42}$$

In the sums paper the convention is

$$(1 + z)^{N-j}(1 - rz)^j = \sum_{n=0}^{N} z^n \phi_{nj}^N, \qquad (1 + z)^{N-j}(1 - z)^j = \sum_{n=0}^{N} z^n \Phi_{nj}^N.$$

Thus the lowercase ϕ denotes general parameters, and uppercase Φ denotes the symmetric specialization in the sums convention. The precise bridge, with the parameters of the transform paper, is

$$r = \frac{p}{q}, \qquad z = 2qv, \qquad K_n(N - 2j, N) = n!(2q)^n \phi_{nj}^N. \tag{43}$$

Parameter warning. The sums paper writes $r = q_S/p_S$. Consequently $p_S = q$ and $q_S = p$ when matching it to the transform paper. Identifying the two letters p without this interchange would give an incorrect biased-parameter dictionary. At $p = q = 1/2$, $K_n(N - 2j, N) = n!\Phi_{nj}^N$. The earlier chapters' ordinary-coefficient notation $K_n(j; N)$ equals Φ_{nj}^N in that symmetric setting; it is not the factorial-normalized $K_n(x, N)$ of this chapter.

2. Orthogonality and inversion

Let J have probabilities $\binom{N}{j} q^j p^{N-j}$ and $X = N - 2J$. Independence of the N Bernoulli steps gives

$$\mathbb{E}[G(v)G(w)] = \{p(1+2qv)(1+2qw)+q(1-2pv)(1-2pw)\}^N = (1+4pqvw)^N.$$

Comparing coefficients proves

$$\mathbb{E}[K_n(X,N)K_m(X,N)] = \delta_{nm}(n!)^2\binom{N}{n}(4pq)^n. \tag{44}$$

To distinguish ordinary from exponential coefficients, put $\widehat{\Phi}_{nj} = K_n(N-2j,N)/n!$ and

$$B = \operatorname{diag}\left(\binom{N}{j}p^{N-j}q^j\right), \qquad \Gamma = \operatorname{diag}\left(\binom{N}{n}(4pq)^n\right).$$

These local source-style matrices are not the unweighted B_N of the symmetric chapters. Orthogonality and its inverse formula become

$$\widehat{\Phi}B\widehat{\Phi}^{\mathsf{T}} = \Gamma, \qquad \widehat{\Phi}^{-1} = B\widehat{\Phi}^{\mathsf{T}}\Gamma^{-1}.$$

Indeed the first identity gives a right inverse of a square matrix. In particular, this binomially weighted row orthogonality must not be confused with the unweighted column sums in the sums chapter.

3. Operator coefficients and positivity

Write $D = d/dx$. For $\deg f \le N$ there is a unique expansion

$$f(x) = \sum_{n=0}^{N} \widetilde{f}(n)K_n(x,N), \qquad \widetilde{f}(n) = \frac{1}{n!}\left[(pe^D + qe^{-D})^{N-n}(\sinh D)^n f\right]_{x=0}.$$

All power series act finitely on the polynomial space. To prove the formula, put $M(z) = pe^z + qe^{-z}$ and $V(z) = \sinh z/M(z)$. Substitution in the generating function gives

$$e^{zx} = M(z)^N \sum_{n\ge 0} \frac{V(z)^n}{n!}K_n(x,N).$$

For coefficient extraction through degree N, only $n \le N$ is needed; compare coefficients of z^m for $0 \le m \le N$ to obtain the claimed operator formula on monomials and hence on every such f.

Proposition 20.1. *If $p \ge q$ and f has nonnegative monomial coefficients, then $\widetilde{f}(n) \ge 0$ for every n. If $p > q$ and every monomial coefficient of f through degree N is strictly positive, then every $\widetilde{f}(n)$ is strictly positive.*

Proof. The series $M(z) = \cosh z + (p-q)\sinh z$ and $\sinh z$ have nonnegative coefficients. If $h(z) = \sum h_m z^m$ and $f(x) = \sum a_m x^m$, then $[h(D)f]_{x=0} = \sum m!h_m a_m$. This proves nonnegativity. For strict positivity the coefficient of z^n in $M(z)^{N-n}(\sinh z)^n$ is 1, so the term a_n already suffices. □

Strict positivity cannot be asserted for arbitrary polynomials with only some positive coefficients: $f = 1$ has zero transform coefficients for $n > 0$. At $p = q$, parity also forces zeros for many individual monomials.

4. Low-degree checks and connection to sums

Set $a = p - q$. Expansion of the generating function gives

$$K_0 = 1, \qquad K_1 = x - Na, \qquad K_2 = (x - Na)^2 + 2a(x - Na) - 4Npq.$$

Consequently

$$x = K_1 + NaK_0, \qquad x^2 = K_2 + 2a(N-1)K_1 + (N^2a^2 + 4Npq)K_0.$$

These examples exhibit positivity when $p \geq q$ and recover $x^2 = K_2 + NK_0$ in the symmetric case.

The weighted partial column sums, Catalan evaluations, and Boolean-lattice applications of `KravSums.tex` are already developed in the sums section included in the existing algebra chapter. Their entries should be read as ϕ^N_{nj} (general) and Φ^N_{nj} (symmetric). For example the bridge for any scalar weights w_n is

$$\sum_{n=0}^{m} w_n(\phi^N_{nj})^2 = \sum_{n=0}^{m} \frac{w_n}{(n!)^2(2q)^{2n}} K_n(N-2j, N)^2.$$

This termwise conversion preserves the weights; replacing only the polynomial symbol would not. It connects the column identities to the operator transform without duplicating the existing sums treatment.

5. Scope of this integration

The supplied transform paper also treats coding theory, modified inversion, and multivariate symmetric representations. This chapter integrates its normalization, orthogonality, expansion calculus, positivity, and their connection with the sums results. The remaining applications are retained in the original reference source and are not claimed to be fully integrated here. The malformed intermediate displays in the supplied Appell-inverse calculation are not copied: the valid identity used above is $V(z) = \sinh z/(pe^z + qe^{-z})$.

APPENDIX A

Tables of Krawtchouk and Symmetric Matrices

1. Verified tables for $N = 2, 3, 4, 5, 7, 8$

The matrices in this section use the standard column-vector basis and zero-based indexing. Each order N gives matrices of size $(N+1) \times (N+1)$:

$$R_N e_j = (N-j)e_{j+1}, \qquad L_N e_j = j e_{j-1}, \qquad A_N = R_N + L_N,$$

$$(\Phi_N)_{nj} = [v^n](1+v)^{N-j}(1-v)^j, \qquad S_N = \Phi_N \operatorname{diag}\binom{N}{j}.$$

For every displayed order, exact integer arithmetic verified

$$[L_N, R_N] = \Lambda_N, \quad A_N\Phi_N = \Phi_N\Lambda_N, \quad \Phi_N A_N = \Lambda_N\Phi_N, \quad \Phi_N^2 = 2^N I, \quad S_N^T = S_N,$$

where $\Lambda_N = \operatorname{diag}(N, N-2, \ldots, -N)$. Definitions and interpretation appear in Sections 1, 1.1, 3.1, and 1.

1.1. $N = 2$.

$$A_2 = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 2 \\ 0 & 1 & 0 \end{pmatrix}.$$

$$R_2 = \begin{pmatrix} 0 & 0 & 0 \\ 2 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix}.$$

$$L_2 = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 2 \\ 0 & 0 & 0 \end{pmatrix}.$$

$$\Phi_2 = \begin{pmatrix} 1 & 1 & 1 \\ 2 & 0 & -2 \\ 1 & -1 & 1 \end{pmatrix}.$$

$$S_2 = \begin{pmatrix} 1 & 2 & 1 \\ 2 & 0 & -2 \\ 1 & -2 & 1 \end{pmatrix}.$$

1.2. $N = 3$**.**

$$A_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 3 & 0 & 2 & 0 \\ 0 & 2 & 0 & 3 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$R_3 = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 3 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$L_3 = \begin{pmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$\Phi_3 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 3 & 1 & -1 & -3 \\ 3 & -1 & -1 & 3 \\ 1 & -1 & 1 & -1 \end{pmatrix}.$$

$$S_3 = \begin{pmatrix} 1 & 3 & 3 & 1 \\ 3 & 3 & -3 & -3 \\ 3 & -3 & -3 & 3 \\ 1 & -3 & 3 & -1 \end{pmatrix}.$$

$$\Lambda_3 = \begin{pmatrix} 3 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & -3 \end{pmatrix}.$$

1.3. $N = 4$**.**

$$A_4 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 4 & 0 & 2 & 0 & 0 \\ 0 & 3 & 0 & 3 & 0 \\ 0 & 0 & 2 & 0 & 4 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$R_4 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ 4 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$L_4 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$\Phi_4 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 4 & 2 & 0 & -2 & -4 \\ 6 & 0 & -2 & 0 & 6 \\ 4 & -2 & 0 & 2 & -4 \\ 1 & -1 & 1 & -1 & 1 \end{pmatrix}.$$

$$S_4 = \begin{pmatrix} 1 & 4 & 6 & 4 & 1 \\ 4 & 8 & 0 & -8 & -4 \\ 6 & 0 & -12 & 0 & 6 \\ 4 & -8 & 0 & 8 & -4 \\ 1 & -4 & 6 & -4 & 1 \end{pmatrix}.$$

1.4. $N = 5$**.**

$$A_5 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 5 & 0 & 2 & 0 & 0 & 0 \\ 0 & 4 & 0 & 3 & 0 & 0 \\ 0 & 0 & 3 & 0 & 4 & 0 \\ 0 & 0 & 0 & 2 & 0 & 5 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$R_5 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 \\ 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$L_5 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$\Phi_5 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 5 & 3 & 1 & -1 & -3 & -5 \\ 10 & 2 & -2 & -2 & 2 & 10 \\ 10 & -2 & -2 & 2 & 2 & -10 \\ 5 & -3 & 1 & 1 & -3 & 5 \\ 1 & -1 & 1 & -1 & 1 & -1 \end{pmatrix}.$$

$$S_5 = \begin{pmatrix} 1 & 5 & 10 & 10 & 5 & 1 \\ 5 & 15 & 10 & -10 & -15 & -5 \\ 10 & 10 & -20 & -20 & 10 & 10 \\ 10 & -10 & -20 & 20 & 10 & -10 \\ 5 & -15 & 10 & 10 & -15 & 5 \\ 1 & -5 & 10 & -10 & 5 & -1 \end{pmatrix}.$$

$$\Lambda_5 = \begin{pmatrix} 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 0 & -3 & 0 \\ 0 & 0 & 0 & 0 & 0 & -5 \end{pmatrix}.$$

1.5. $N = 7$**.**

$$A_7 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$R_7 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$L_7 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 7 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$\Phi_7 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 7 & 5 & 3 & 1 & -1 & -3 & -5 & -7 \\ 21 & 9 & 1 & -3 & -3 & 1 & 9 & 21 \\ 35 & 5 & -5 & -3 & 3 & 5 & -5 & -35 \\ 35 & -5 & -5 & 3 & 3 & -5 & -5 & 35 \\ 21 & -9 & 1 & 3 & -3 & -1 & 9 & -21 \\ 7 & -5 & 3 & -1 & -1 & 3 & -5 & 7 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 \end{pmatrix}.$$

$$S_7 = \begin{pmatrix} 1 & 7 & 21 & 35 & 35 & 21 & 7 & 1 \\ 7 & 35 & 63 & 35 & -35 & -63 & -35 & -7 \\ 21 & 63 & 21 & -105 & -105 & 21 & 63 & 21 \\ 35 & 35 & -105 & -105 & 105 & 105 & -35 & -35 \\ 35 & -35 & -105 & 105 & 105 & -105 & -35 & 35 \\ 21 & -63 & 21 & 105 & -105 & -21 & 63 & -21 \\ 7 & -35 & 63 & -35 & -35 & 63 & -35 & 7 \\ 1 & -7 & 21 & -35 & 35 & -21 & 7 & -1 \end{pmatrix}.$$

1.6. $N = 8$**.**

$$A_8 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 8 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$R_8 = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 8 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 7 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 6 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 5 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 3 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \end{pmatrix}.$$

$$L_8 = \begin{pmatrix} 0 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 3 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 4 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 6 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 7 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{pmatrix}.$$

$$\Phi_8 = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 8 & 6 & 4 & 2 & 0 & -2 & -4 & -6 & -8 \\ 28 & 14 & 4 & -2 & -4 & -2 & 4 & 14 & 28 \\ 56 & 14 & -4 & -6 & 0 & 6 & 4 & -14 & -56 \\ 70 & 0 & -10 & 0 & 6 & 0 & -10 & 0 & 70 \\ 56 & -14 & -4 & 6 & 0 & -6 & 4 & 14 & -56 \\ 28 & -14 & 4 & 2 & -4 & 2 & 4 & -14 & 28 \\ 8 & -6 & 4 & -2 & 0 & 2 & -4 & 6 & -8 \\ 1 & -1 & 1 & -1 & 1 & -1 & 1 & -1 & 1 \end{pmatrix}.$$

$$S_8 = \begin{pmatrix} 1 & 8 & 28 & 56 & 70 & 56 & 28 & 8 & 1 \\ 8 & 48 & 112 & 112 & 0 & -112 & -112 & -48 & -8 \\ 28 & 112 & 112 & -112 & -280 & -112 & 112 & 112 & 28 \\ 56 & 112 & -112 & -336 & 0 & 336 & 112 & -112 & -56 \\ 70 & 0 & -280 & 0 & 420 & 0 & -280 & 0 & 70 \\ 56 & -112 & -112 & 336 & 0 & -336 & 112 & 112 & -56 \\ 28 & -112 & 112 & 112 & -280 & 112 & 112 & -112 & 28 \\ 8 & -48 & 112 & -112 & 0 & 112 & -112 & 48 & -8 \\ 1 & -8 & 28 & -56 & 70 & -56 & 28 & -8 & 1 \end{pmatrix}.$$

2. Additional orders through $N = 14$

ADDITIONAL ORDERS THROUGH $N = 14$ NEEDS TO BE DONE.

The verified tables above are the currently developed subset of the original program for orders one through fourteen. Additional orders have not been filled with unverified or placeholder numerical values.

APPENDIX B

Polynomial Tables

The developed tables cover $N = 3, 5$, with every degree $0 \le n \le N$. The normalization is

$$(1+v)^{N-j}(1-v)^j = \sum_{n=0}^{N} K_n(j;N)v^n, \qquad k_n(N-2j,N) = n!K_n(j;N).$$

All lattice entries have been verified exactly against the binomial coefficient formula and Appendix A.

1. Centered variables x and N

From $k_0 = 1$, $k_1 = x$ and $k_{n+1} = xk_n - n(N-n+1)k_{n-1}$, we obtain

$$\begin{aligned}
k_0(x,N) &= 1\\
k_1(x,N) &= x\\
k_2(x,N) &= -N + x^2\\
k_3(x,N) &= -3Nx + x^3 + 2x\\
k_4(x,N) &= 3N^2 - 6Nx^2 - 6N + x^4 + 8x^2\\
k_5(x,N) &= 15N^2x - 10Nx^3 - 50Nx + x^5 + 20x^3 + 24x
\end{aligned}$$

1.1. $N = 3$**.** The lattice is $x = 3, 1, -1, -3$.

$$\begin{aligned}
k_0(x,3) &= 1\\
k_1(x,3) &= x\\
k_2(x,3) &= x^2 - 3\\
k_3(x,3) &= x^3 - 7x
\end{aligned}$$

1.2. $N = 5$**.** The lattice is $x = 5, 3, 1, -1, -3, -5$.

$$\begin{aligned}
k_0(x,5) &= 1\\
k_1(x,5) &= x\\
k_2(x,5) &= x^2 - 5\\
k_3(x,5) &= x^3 - 13x\\
k_4(x,5) &= x^4 - 22x^2 + 45\\
k_5(x,5) &= x^5 - 30x^3 + 149x
\end{aligned}$$

2. Degree and lattice variables i and j

Rows are degrees $i = 0, \ldots, N$ and columns are lattice indices $j = 0, \ldots, N$. These are Φ_N matrices of $K_i(j; N)$ values, not factorial-scaled k_i values.

$$\big(K_i(j;3)\big)_{i,j=0}^{3} = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 3 & 1 & -1 & -3 \\ 3 & -1 & -1 & 3 \\ 1 & -1 & 1 & -1 \end{pmatrix}.$$

$$\big(K_i(j;5)\big)_{i,j=0}^{5} = \begin{pmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 5 & 3 & 1 & -1 & -3 & -5 \\ 10 & 2 & -2 & -2 & 2 & 10 \\ 10 & -2 & -2 & 2 & 2 & -10 \\ 5 & -3 & 1 & 1 & -3 & 5 \\ 1 & -1 & 1 & -1 & 1 & -1 \end{pmatrix}.$$

3. Lattice variables j and N; orders 1 through 20

The following formulas supply degrees zero through five. For fixed N, retain only $n \le N$. The orders completed here are $N = 3, 5$; the other orders in the original program remain to be developed.

$$K_0(j;N) = 1$$
$$K_1(j;N) = N - 2j$$
$$K_2(j;N) = \frac{N^2 - 4Nj - N + 4j^2}{2}$$
$$K_3(j;N) = -\frac{(-N+2j)\left(N^2 - 4Nj - 3N + 4j^2 + 2\right)}{6}$$
$$K_4(j;N) = \frac{N^4 - 8N^3j - 6N^3 + 24N^2j^2 + 24N^2j + 11N^2 - 32Nj^3 - 24Nj^2 - 32Nj - 6N + 16j^4 + 3}{24}$$
$$K_5(j;N) = -\frac{(-N+2j)\left(N^4 - 8N^3j - 10N^3 + 24N^2j^2 + 40N^2j + 35N^2 - 32Nj^3 - 40Nj^2 - 80Nj -}{120}$$

3.1. $N = 3$.

$$K_0(j;3) = 1$$
$$K_1(j;3) = 3 - 2j$$
$$K_2(j;3) = 2j^2 - 6j + 3$$
$$K_3(j;3) = -\frac{(2j-3)\left(2j^2 - 6j + 1\right)}{3}$$

3.2. $N = 5$**.**

$$K_0(j;5) = 1$$
$$K_1(j;5) = 5 - 2j$$
$$K_2(j;5) = 2\left(j^2 - 5j + 5\right)$$
$$K_3(j;5) = -\frac{2\,(2j-5)\left(j^2 - 5j + 3\right)}{3}$$
$$K_4(j;5) = \frac{2j^4 - 20j^3 + 64j^2 - 70j + 15}{3}$$
$$K_5(j;5) = -\frac{(2j-5)\left(2j^4 - 20j^3 + 60j^2 - 50j + 3\right)}{15}$$

APPENDIX C

Spectral Tables

1. Eigenvalues of S

EIGENVALUES OF S NEEDS TO BE DONE.

2. Normalization and numerical verification

NORMALIZATION AND NUMERICAL VERIFICATION NEEDS TO BE DONE.

APPENDIX D

Multivariate Extensions

1. Remarks on the multivariate case

REMARKS ON THE MULTIVARIATE CASE NEEDS TO BE DONE.

APPENDIX E

Time-Zero Polynomial Tables

1. Definitions and tables

DEFINITIONS AND TABLES NEEDS TO BE DONE.

APPENDIX F

Mikhail Kravchuk: A Biographical Sketch

1. Biographical chronology

BIOGRAPHICAL CHRONOLOGY NEEDS TO BE DONE.

2. Primary sources and historical bibliography

PRIMARY SOURCES AND HISTORICAL BIBLIOGRAPHY NEEDS TO BE DONE.

Bibliography and Source Register

Supplied source materials

- `sources/KravFeinsKocik.tex`: the uploaded article source supplying the encyclopedia outline in Section 6.1.
- `sources/jkhadamard_Oct_08_00.tex`: the uploaded Hadamard, tensor, and random-walk article source.
- `Kravchuk.zip`: historical website source supplying additional headings and the introductory chapter. This original archive is not present under that filename in the active source package; the extracted heading inventory and prior project archives preserve the available local provenance.
- `KravConvolutions.pdf`: supporting material named in the earlier project register, but not present in the active project. No new integration or claim of source inspection is made for this unavailable file.
- `sources/KravEstimate.tex`: the newly supplied absolute-entry-sum estimate note, retained unchanged; selected results and corrections are developed in Section 6. The retained local sums manuscripts in `supporting-papers/` supply further selected formulas with proofs in that section.
- `sources/KrawtchoukFeynman.pdf`: the locally retained article underlying the integrated `KocikFeynman` discussions; normalization and source corrections are recorded in the integration notes.
- `KrawtchoukReferences_bibliography.tex`: the supplied list of fifty references, archived unchanged in `sources/`.

Integration guide for the two uploaded TeX sources

Both original files are preserved byte-for-byte in `sources/`. They are standalone article documents, not chapter fragments, and are therefore not directly input into the book.

- `KravFeinsKocik.tex`: matrix definitions and Hadamard construction connect to Chapters 14, 15, and 10; symmetric tensors to Chapter 9; the Ehrenfest model to Chapter 4; classical random walks to Chapter 3; its product-martingale and finite-valued multinomial construction is developed in Chapter 6; and Section 6.1 supplies the existing Encyclopedia program.

- `jkhadamard_Oct_08_00.tex`: matrix definitions and identities connect to Chapters 14 and 17; the Hadamard construction and quantum random walk to Chapter 10; classical random walks to Chapter 3; and tensor appendices to Chapter 9.

The basis-explicit Hadamard compression is developed in Section 7. Source filenames and section headings identify these local documents; publication metadata and the articles' internal reference lists have not been independently verified or merged into the literature bibliography.

Current integration and verification status

The supplied fifty literature records are retained as a reference collection, not as fifty verified sources for the developed chapters. At this audit, none of the keys `ref01`–`ref50` is cited in the running text. No citations have been assigned merely because a title sounds relevant. Bibliographic wording has been preserved; independent metadata review and claim-level integration remain pending.

The two supplied standalone TeX articles are linked by filename and local section descriptions rather than being directly input, so their separate preambles and bibliography keys do not collide with the book. The integrated article cited as `KocikFeynman` has an explicit notation dictionary in the front matter. The martingale chapter now links to the finite-valued construction in the supplied source. Gaussian summation and the terminal-polynomial remark are self-contained derivations from the established normalization, with reproducible numerical data rather than an unsupported literature attribution.

For the review inventory, see `GAP_REVIEW.md` and `sources/INTEGRATION_AUDIT.md`. The moments-updated manuscript and the previously named convolution PDF are not available in the active project and have not been reconstructed. The supplied `main.pdf` predates the recent edits; compile `main.tex` to review the current book.

The supplied `KravEstimate.tex` is now preserved unchanged in `sources/`. Section 6 integrates checked estimates and selected exact sums from the retained local sums manuscripts, explicitly correcting the missing summation and near-boundary factor two. Details and verification scope are recorded in `sources/INTEGRATION_KravEstimate.md`; the literature bibliography is unchanged.

Literature bibliography

The supplied list is incorporated below with its original citation keys `ref01`–`ref50` and bibliographic wording preserved. These records are user-supplied and have not been independently validated in this integration; the separate review file records items requiring attention. The previously verified integrated-article entry `KocikFeynman` is retained.

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